

IRIS FLOWER CLASSIFICATION



BY: MUHAMMAD ANAS RATHORE

ACKNOWLEDGEMENT

*First and foremost, all praise and gratitude are due to **Allah SWT**, whose guidance, blessings, and mercy enabled me to complete this project successfully. Without **His** support, none of this work would have been possible.*

*I would like to express my deepest gratitude to my **parents**, whose unwavering love, encouragement, and sacrifices have been a constant source of strength and motivation throughout my academic journey. Their belief in my abilities has inspired me to strive for excellence in every endeavor.*

*I would also like to sincerely acknowledge **Miss Mahrose Fatima Naqvi, Asma Yahya, and Asma Qaiser**, my mentors, for being inspiring guides during my studies at **Iqra University**. Their dedication to teaching and academic guidance has positively influenced my learning journey.*

Finally, I am grateful to all my friends, colleagues, and everyone who directly or indirectly contributed to the successful completion of this project. Their encouragement and support made this journey smoother and more enjoyable.

This work stands because of the guidance, support, and inspiration I received from all the above-mentioned individuals, and I am sincerely thankful to each of them.

TABLE OF CONTENT

<i>Problem Statement</i>	<i>4</i>
<i>Objective</i>	<i>5</i>
<i>Data Description</i>	<i>6</i>
<i>Data Preprocessing</i>	<i>7</i>
<i>Exploratory Data Analysis</i>	<i>8</i>
<i>Flow Chart</i>	<i>10</i>
<i>Feature Engineering</i>	<i>11</i>
<i>Train Test Split</i>	<i>12</i>
<i>Machine Learning Models</i>	<i>13</i>
<i>Model Evaluation Metrics</i>	<i>14</i>
<i>Model Performance Comparison</i>	<i>15</i>
<i>Feature Importance Analysis</i>	<i>16</i>
<i>Final Model Selection</i>	<i>17</i>
<i>Outputs, Graphs and Visualization</i>	<i>18</i>
<i>Conclusion</i>	<i>49</i>
<i>Writer's Introduction</i>	<i>50</i>

Problem Statement

Classification is one of the most important applications of machine learning, where a model learns patterns from historical data and predicts the correct category of new observations. The Iris Flower dataset is a widely used benchmark dataset that contains measurements of iris flowers belonging to three species: Iris Setosa, Iris Versicolor, and Iris Virginica.

The primary challenge is to accurately classify an iris flower based on its physical characteristics, including sepal length, sepal width, petal length, and petal width. Although the dataset is relatively small and structured, different machine learning algorithms may produce different levels of classification performance due to their unique learning mechanisms.

This project focuses on developing an automated classification system by analyzing the Iris dataset and applying multiple machine learning algorithms. The study aims to:

- Understand the patterns and relationships present in the dataset.
- Identify the most significant features contributing to species classification.
- Train and evaluate multiple machine learning models.
- Compare model performance using standard evaluation metrics.
- Select the most accurate and reliable model for iris species prediction.

To achieve this, several classification algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Support Vector Machine (SVM), and Naïve Bayes are implemented and compared. The final objective is to determine the best-performing model capable of accurately classifying iris flower species while demonstrating the effectiveness of machine learning techniques for multiclass classification problems.

Objective

The Iris Flower Classification project aims to develop a machine learning-based system capable of accurately identifying iris flower species using their physical measurements. The project involves data preprocessing, exploratory analysis, model development, and performance evaluation to determine the most effective classification algorithm. By comparing multiple machine learning techniques, the study seeks to identify the model that provides the highest prediction accuracy and reliability.

The objectives of this project are:

1. To perform data preprocessing on the Iris dataset.
2. To conduct Exploratory Data Analysis (EDA) to understand data distribution and patterns.
3. To apply feature engineering and feature scaling techniques for improved model performance.
4. To train multiple machine learning classification models on the dataset.
5. To evaluate and compare model performance using standard evaluation metrics.
6. To analyze the importance and contribution of individual features in classification.
7. To identify the best-performing classification model based on experimental results.
8. To predict the species of new iris flower samples using the selected model.
9. To demonstrate the effectiveness of machine learning techniques for multiclass classification problems.

Dataset Description

The Iris Flower Dataset was used in this project to develop and evaluate multiple machine learning classification models. It is one of the most popular benchmark datasets for classification tasks and contains measurements of iris flowers belonging to three different species. The dataset provides a suitable environment for comparing the performance of various machine learning algorithms due to its balanced classes and well-defined numerical features.

Dataset Overview

- The dataset was loaded using the Pandas library.
- It contains measurements of iris flowers from three species: Setosa, Versicolor, and Virginica.
- The target variable is **Species**, while the remaining attributes are numerical input features.
- Each record represents a single flower sample.

Head Values

- The head() function was used to display the first five records of the dataset.
- The output showed sample values for sepal length, sepal width, petal length, petal width, and species.

Tail Values

- The tail() function was used to display the last five records of the dataset.
- This provided a quick inspection of the dataset's ending entries. It confirmed the consistency of data formatting throughout the dataset.

Dataset Information

- The info() function was used to obtain detailed information about the dataset.
- All feature columns were numerical, while the species column was categorical before encoding.
- No missing values were observed in any column.

Shape

- The shape attribute was used to determine the dimensions of the dataset.
- The dataset initially contained 150 rows and 5 columns.
- Four columns represented input features, while one column represented the target class.

Overall, the dataset was found to be clean, structured, and suitable for machine learning classification tasks after basic preprocessing and duplicate removal.

Data Preprocessing

Missing Value Analysis

- The `isnull().sum()` function was applied to check for missing values.
- No missing values were found in the dataset.
- This indicated that the dataset was complete and ready for further analysis.

Duplicate Value Analysis

- The `duplicated().sum()` function was used to identify duplicate entries.
- A small number of duplicate records were detected.
- These duplicate rows were removed using the `drop_duplicates()` function to improve data quality and model reliability.

Statistical Summary

- The `describe()` function was used to generate descriptive statistics of the numerical features.
- Measures such as mean, standard deviation, minimum value, maximum value, and quartiles were obtained.
- The statistical summary provided insights into the distribution and spread of each feature.

Label Encoding

- The target variable (**Species**) was originally stored in categorical text format.
- Machine learning algorithms require numerical input; therefore, label encoding was applied.
- The `LabelEncoder()` class was used to transform species names into numerical labels.

Example:

- *Setosa* → 0
- *Versicolor* → 1
- *Virginica* → 2

After completing these preprocessing steps, the dataset became clean, standardized, and ready for exploratory data analysis and machine learning model development.

Exploratory Data Analysis (EDA)

Several exploratory techniques were used to understand the dataset.

Species Distribution

- A count plot was used to visualize the distribution of the target classes in the dataset.
- The dataset is well balanced across all three Iris species.
- This balance ensures that the model does not become biased toward any particular class and supports fair training and evaluation.

Feature Distribution

- Histograms were plotted for all numerical features, including sepal length, sepal width, petal length, and petal width.
- Petal length and petal width show strong separation between species, making them highly useful for classification.
- Sepal-based features show more overlap, indicating lower discriminatory power compared to petal features.

Scatter Plot Analysis

- A scatter plot of petal length versus petal width was used to study the relationship between these two important features.
- The Setosa species forms a distinct cluster and is clearly separable from the other two classes.
- Versicolor and Virginica show partial overlap, which makes them more difficult to distinguish using simple linear separation.

Box Plot Analysis

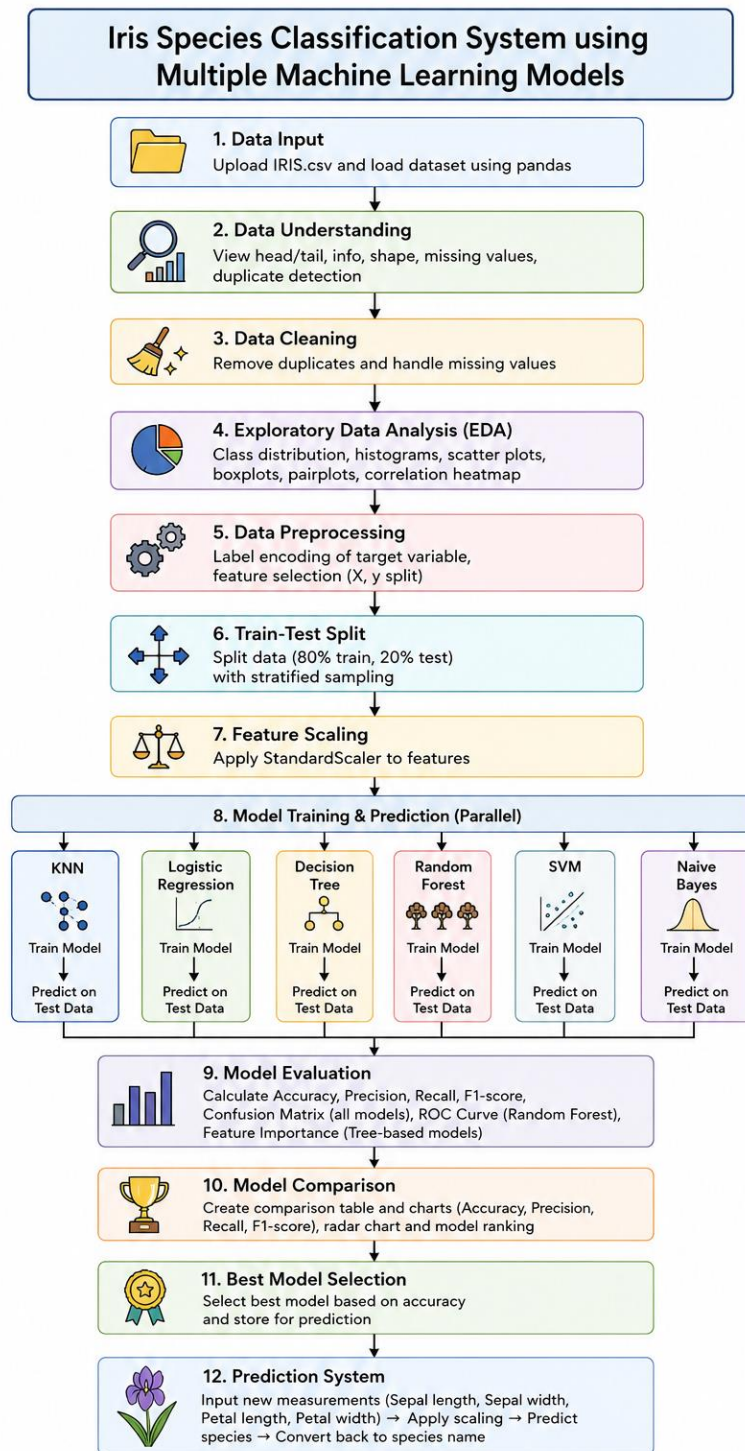
- Boxplots were used to examine the distribution of features and identify possible outliers.
- Most data points fall within a reasonable range, with only a few mild outliers observed.
- These outliers are not extreme and do not significantly affect the overall distribution or model performance.
- A pair plot was generated to visualize pairwise relationships between all numerical features.
- Petal-related features provide strong class separation compared to sepal features.

- The visual patterns confirm that petal length and petal width are the most important features for distinguishing Iris species.

Correlation Matrix

- A heatmap was used to analyze correlations between numerical features.
- Petal length and petal width show a strong positive correlation, meaning they tend to increase together.
- Sepal features show weaker correlations with petal features, indicating they contribute less to class separation compared to petal measurements.

Flow Chart



This image is AI-generated.

Feature Engineering

Feature Selection

1. The *Species* column was selected as the target variable (y), which represents the class labels of the Iris flowers.
2. The remaining four numerical attributes were selected as input features (X) for model training.
3. These features include sepal length, sepal width, petal length, and petal width.
4. These measurements represent the physical characteristics of the flowers and provide sufficient information for distinguishing between different Iris species.
5. Using all four features helps the model learn complex patterns and improves classification accuracy.

Feature Scaling

1. Feature scaling was performed using the **StandardScaler()** technique from the Scikit-learn library.
2. This method standardizes the dataset by transforming feature values to a common scale, ensuring that no feature dominates due to larger numerical values.
3. It is especially important for machine learning algorithms that depend on distance calculations, such as K-Nearest Neighbors (KNN) and Support Vector Machine (SVM).
4. The scaler was first fitted on the training dataset to learn the mean and standard deviation of each feature.
5. The same transformation was then applied to both the training and testing datasets to maintain consistency and prevent data leakage.

The standardization formula is:

$$Z = \frac{X - \mu}{\sigma}$$

Where;

X = Original feature value

μ = Mean of the feature

σ = Standard deviation of the feature

Train-Test Split

The dataset was divided into training and testing sets to evaluate model performance on unseen data. An 80%–20% split was used, where 80% of the data is used for training and 20% is reserved for testing. The split was implemented using the `train_test_split()` function from Scikit-learn.

Benefits of Train-Test Split:

1. Helps prevent overfitting by evaluating the model on unseen data.
2. Provides an unbiased estimate of model performance.
3. Maintains class distribution, improving the reliability of evaluation results.

Machine Learning Models

Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their outputs to produce a final prediction. It improves accuracy and reduces overfitting by using majority voting in classification tasks.

Decision Tree

Decision Tree is a supervised learning algorithm that makes predictions by splitting data into branches based on feature conditions. It creates a tree-like structure of decisions that is easy to interpret but may overfit if not properly controlled.

Naïve Bayes

Naive Bayes is a probabilistic classification algorithm based on Bayes' Theorem that assumes all features are independent of each other. Despite this simplified assumption, it performs efficiently and is widely used in text classification and spam detection tasks.

Logistic Regression

Logistic Regression is a statistical classification algorithm used to predict the probability of categorical outcomes, typically for binary and multi-class problems. It uses the sigmoid function to convert predicted values into probabilities and assigns class labels based on a threshold.

Support Vector Machine (SVM)

SVM (Support Vector Machine) is a supervised learning algorithm that finds the optimal hyperplane to separate different classes with the maximum margin between them. It is highly effective for both linear and non-linear classification problems using different kernel functions.

K-Nearest Neighbors (KNN)

KNN (K-Nearest Neighbors) is a simple supervised learning algorithm that classifies a new data point based on the majority class of its nearest neighbors using distance metrics such as Euclidean distance. It is easy to implement and works well for small datasets, but it can become slow when handling large datasets.

Model Evaluation Metrics

Accuracy

Accuracy is the performance metric that measures the proportion of correctly predicted observations out of the total observations. It shows how often the model is correct overall.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall

Recall is the metric that measures how many actual positive cases were correctly identified by the model. It focuses on the model's ability to find all positive instances.

$$Recall = \frac{TP}{TP + FN}$$

Precision

Precision is the metric that measures how many of the predicted positive cases are actually correct. It focuses on the quality of positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

F1- Score

F1 Score is the harmonic mean of precision and recall. It provides a balanced measure when there is an uneven class distribution and both precision and recall are important.

$$F1\ Score = \frac{2 \times (Precision \times Recall)}{Precision + Recall}$$

Where:

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

Model Performance Comparison

MODELS NAME	ACCURACY	RECALL	PERCISION	F1-SCORE	PERFOMANCE
<i>Logistic Regression</i>	<i>0.9333</i>	<i>0.9333</i>	<i>0.9333</i>	<i>0.9333</i>	<i>Good</i>
<i>K-Nearest Neighbors (KNN)</i>	<i>0.9333</i>	<i>0.9444</i>	<i>0.9333</i>	<i>0.9327</i>	<i>Good</i>
<i>Decision Tree</i>	<i>0.9333</i>	<i>0.9333</i>	<i>0.9333</i>	<i>0.9333</i>	<i>Good</i>
<i>Random Forest</i>	<i>0.9667</i>	<i>0.9697</i>	<i>0.9667</i>	<i>0.9666</i>	<i>Excellent</i>
<i>Support Vector Machine (SVM)</i>	<i>0.9667</i>	<i>0.9697</i>	<i>0.9667</i>	<i>0.9666</i>	<i>Excellent</i>
<i>Naive Bayes</i>	<i>0.9333</i>	<i>0.9333</i>	<i>0.9333</i>	<i>0.9333</i>	<i>Good</i>

The performance evaluation shows that all machine learning models achieved strong classification results on the Iris dataset, with accuracy values greater than 0.93. This indicates that the selected features effectively distinguish between the three Iris species.

Random Forest and Support Vector Machine (SVM) achieved the highest performance, each obtaining an accuracy of **0.9667**, a precision of **0.9697**, a recall of **0.9667**, and an F1-score of **0.9666**. These results demonstrate their ability to classify flower species with very high reliability and consistency.

Logistic Regression, Decision Tree, and Naive Bayes produced identical performance scores, achieving an accuracy, precision, recall, and F1-score of **0.9333**. These models successfully classified most samples and demonstrated stable predictive performance.

K-Nearest Neighbors (KNN) also achieved an accuracy of **0.9333**, with a slightly higher precision value of **0.9444**, indicating strong positive prediction capability. However, its overall performance remained slightly lower than that of Random Forest and SVM.

Based on the evaluation metrics, **Random Forest** and **Support Vector Machine (SVM)** were identified as the best-performing models for the Iris Species Classification. Their superior accuracy and balanced performance across all evaluation measures make them the most suitable choices for the final classification task.

Feature Importance Analysis

Feature importance analysis was performed to identify which features contribute the most to the classification of Iris species. Understanding feature importance helps determine the attributes that have the greatest influence on model predictions and overall classification accuracy.

1. **Petal Length** was identified as one of the most influential features for distinguishing between Iris species.
2. **Petal Width** also showed high importance and played a significant role in the classification process.
3. **Sepal Length** contributed moderately to the prediction of flower species.
4. **Sepal Width** had the lowest importance among the four features due to greater overlap between species.

Overall, petal-related features provided the strongest discriminatory power and were the key factors in achieving high classification accuracy.

Final Model Selection

Based on the performance evaluation results, the final model was selected by comparing the Accuracy, Precision, Recall, and F1-Score of all trained machine learning algorithms. Both **Random Forest** and **Support Vector Machine (SVM)** achieved the highest performance among all models.

1. Random Forest achieved an accuracy of **0.9667**, along with excellent Precision, Recall, and F1-Score values.
2. Support Vector Machine (SVM) also achieved an accuracy of **0.9667**, demonstrating equally strong classification performance.
3. Both models outperformed Logistic Regression, KNN, Decision Tree, and Naive Bayes.
4. Due to its high accuracy, robustness, and ability to handle feature interactions effectively, **SVM was selected as the final model** for the Iris Species Classification System.
5. The selected model was saved and used for predicting the species of new Iris flower samples.

Outputs, Graphs and Visualization

The project includes the following visual analyses:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Chart 1: Head Values of Dataset

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

Chart 2: Tail Values of Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

Chart 3: Dataset Information

IRIS FLOWER CLASSIFICATION

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

Chart 4: Dataset Description



Figure 1: Heatmap Visualization for Clean Dataset

IRIS FLOWER CLASSIFICATION

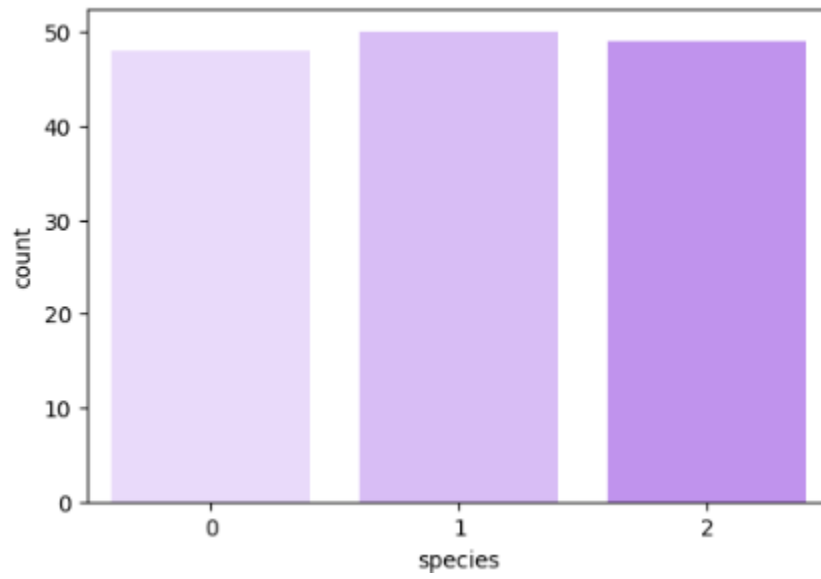


Figure 2: Species Distribution Graph

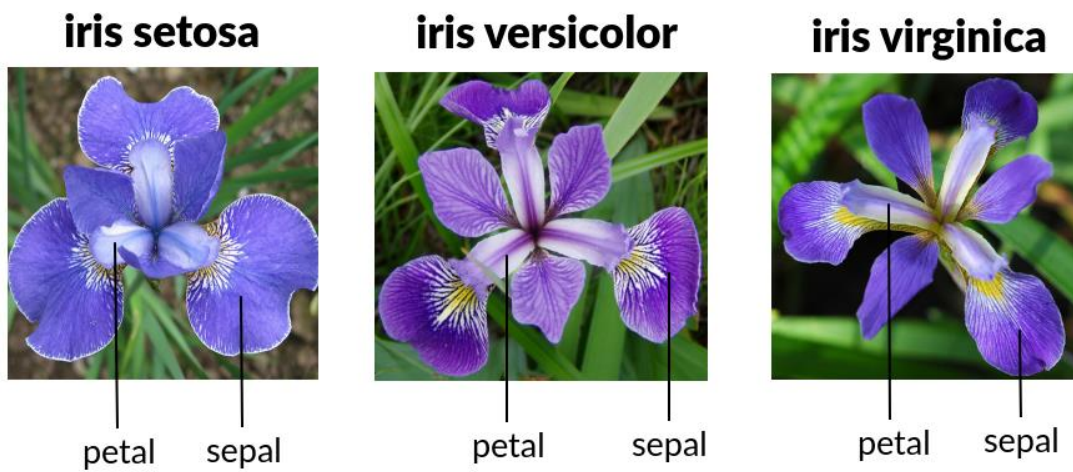


Figure 3: Types of Iris Flower

IRIS FLOWER CLASSIFICATION

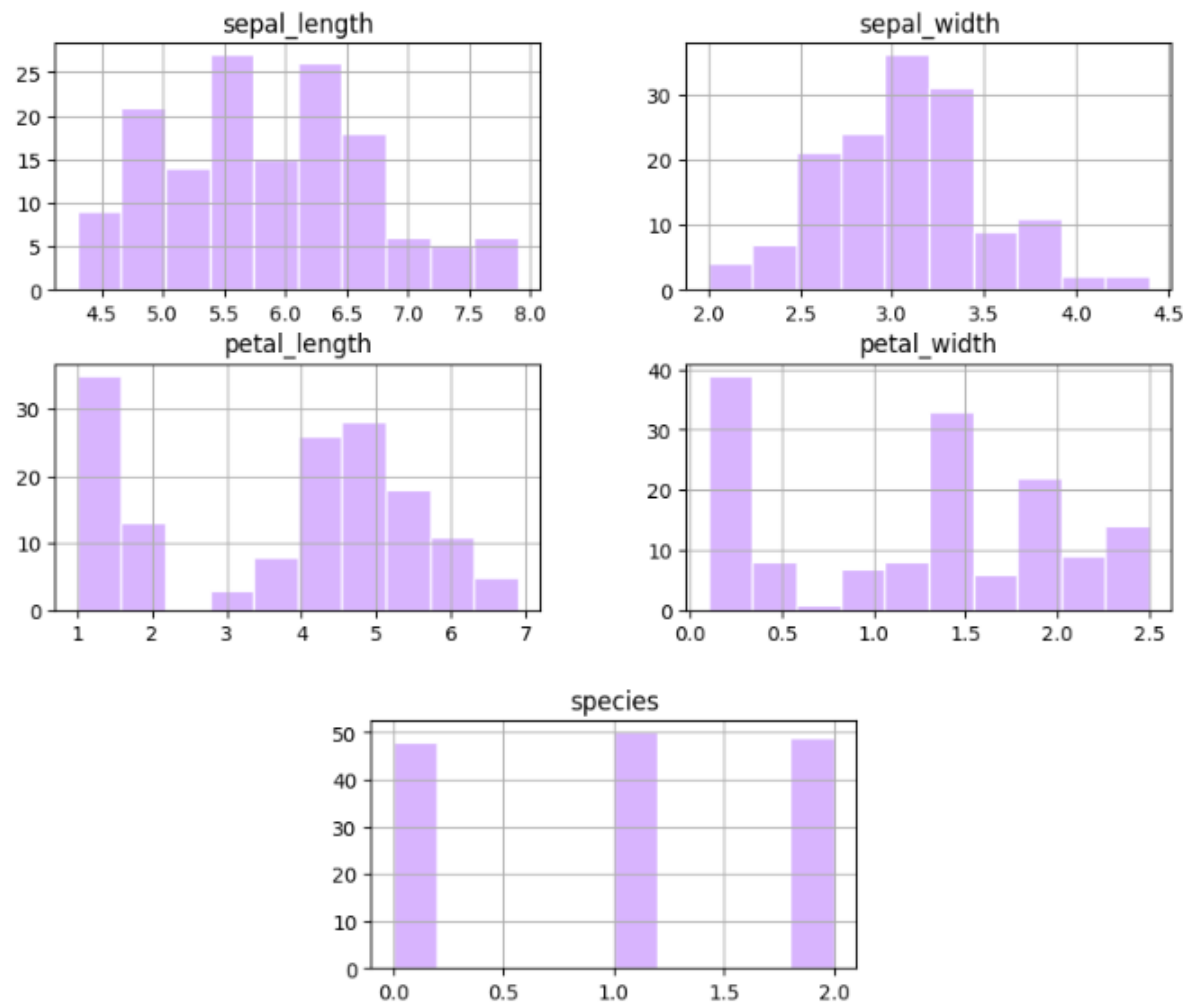


Figure 4: Feature Distribution Histogram

IRIS FLOWER CLASSIFICATION

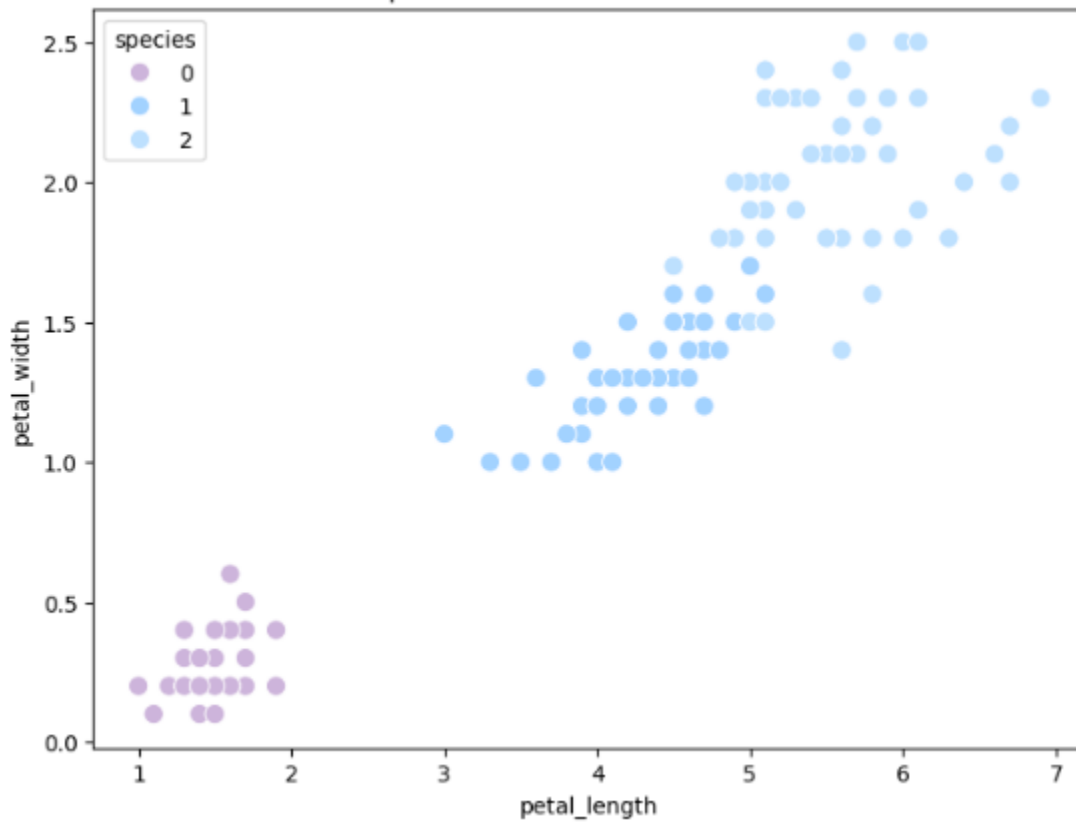


Figure 5: Petal Features Distribution Graph

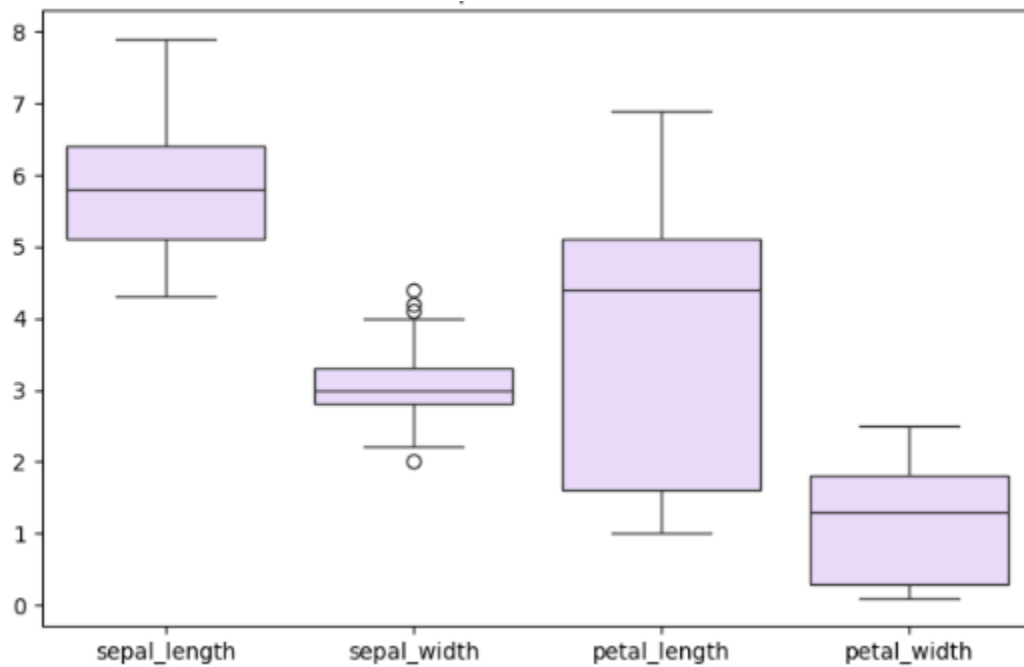


Figure 6: Box Plots for Each Feature

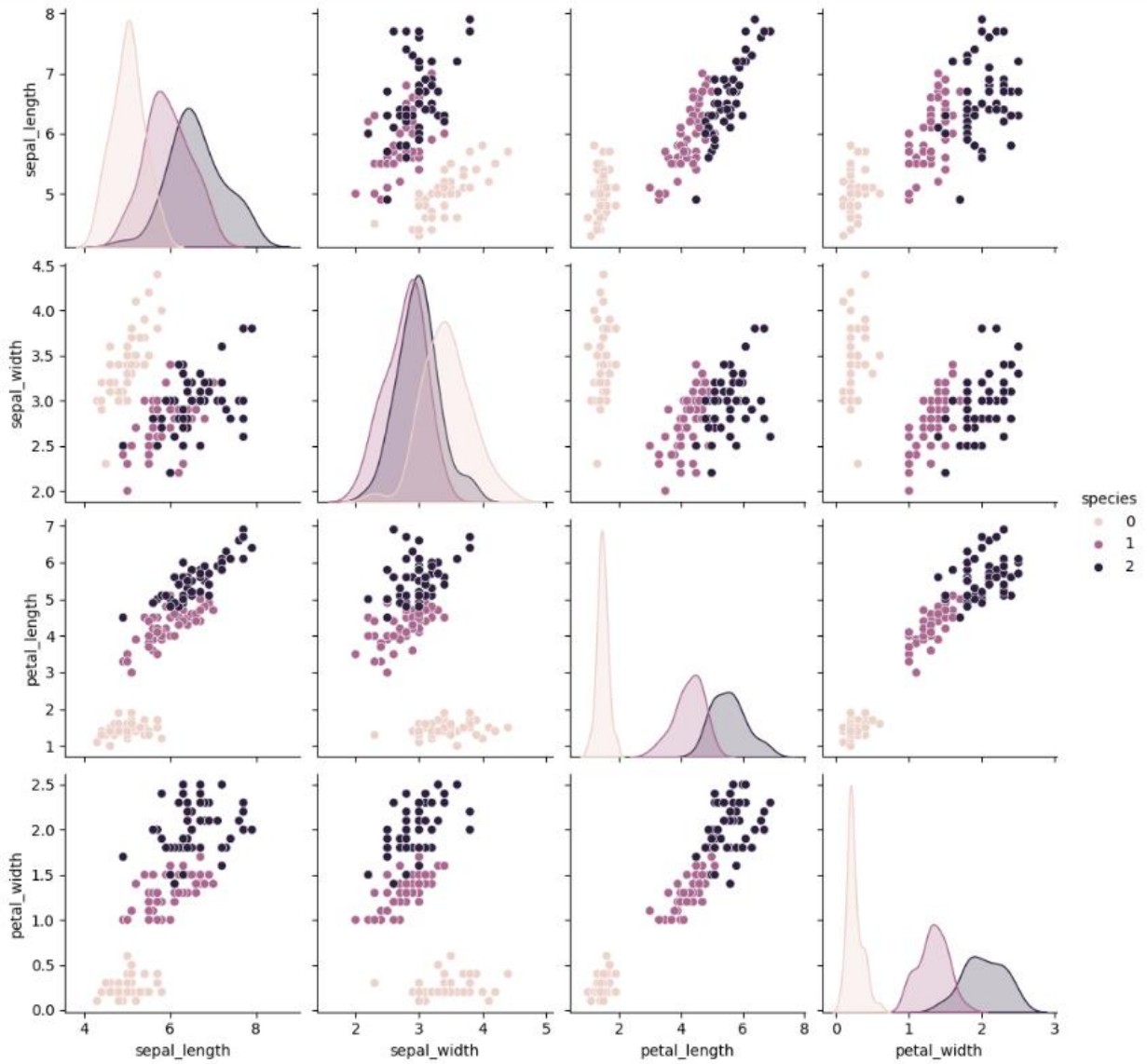


Figure 7: Pair Plot Graph for Each Species and their Features

IRIS FLOWER CLASSIFICATION

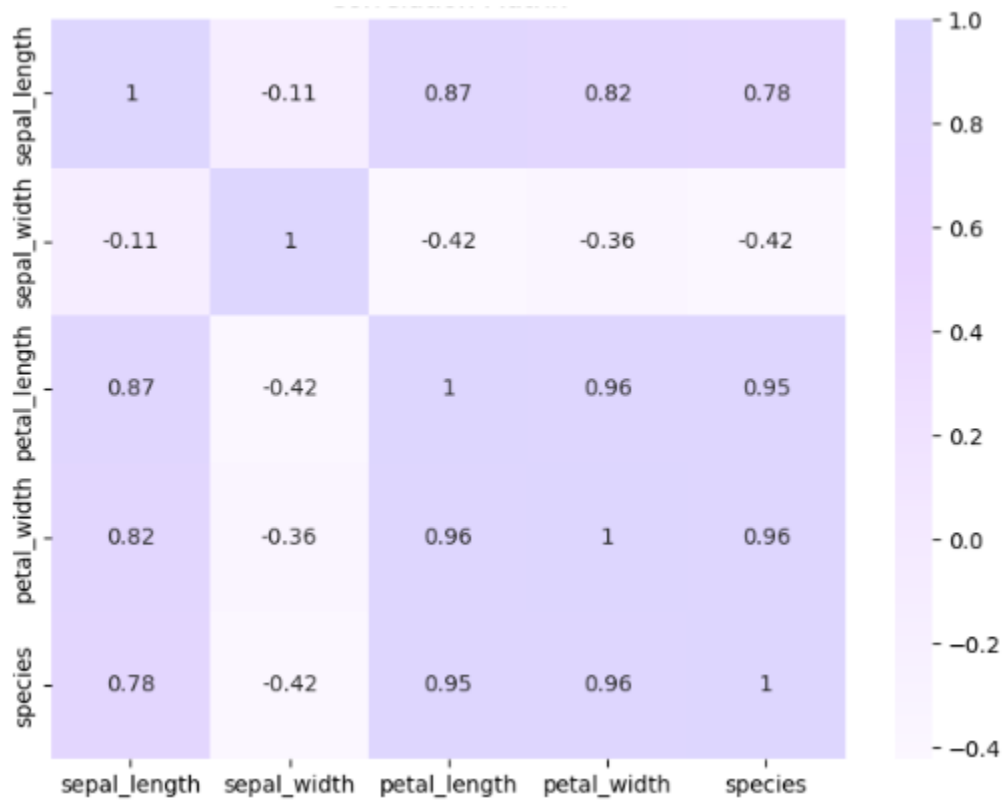


Figure 8: Correlation Matrix

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.933333	0.933333	0.933333	0.933333
1	KNN	0.933333	0.944444	0.933333	0.932660
2	Decision Tree	0.933333	0.933333	0.933333	0.933333
3	Random Forest	0.966667	0.969697	0.966667	0.966583
4	SVM	0.966667	0.969697	0.966667	0.966583
5	Naive Bayes	0.933333	0.933333	0.933333	0.933333

Chart 5: Evaluation of ML Models using different metrics

IRIS FLOWER CLASSIFICATION

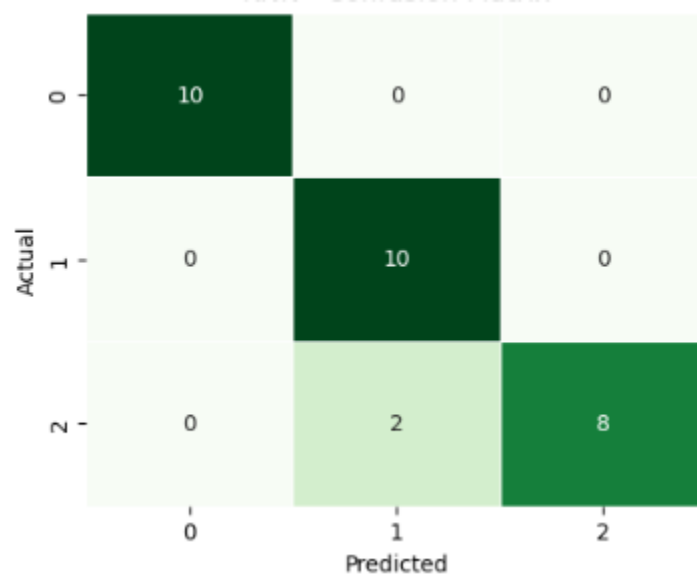


Figure 9: Confusion Matrix, KNN

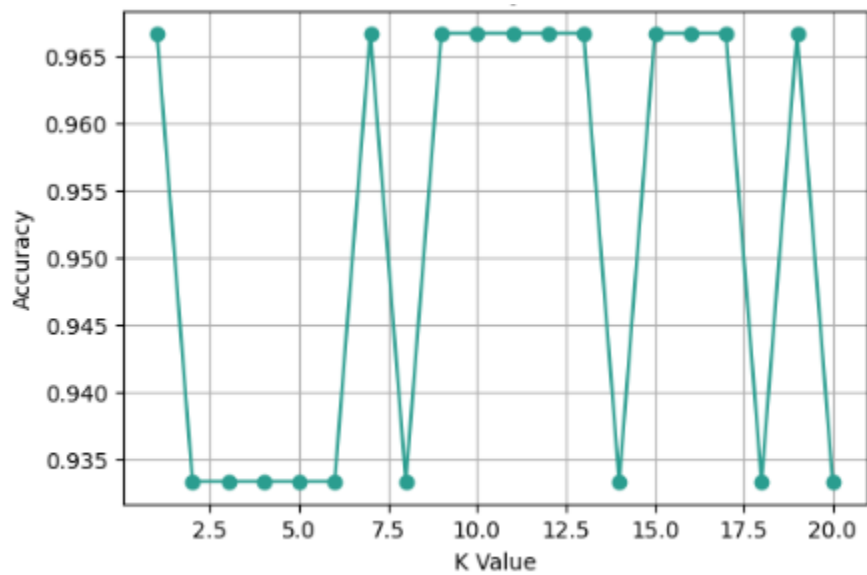


Figure 10: Accuracy Vs K- Value Graph

IRIS FLOWER CLASSIFICATION

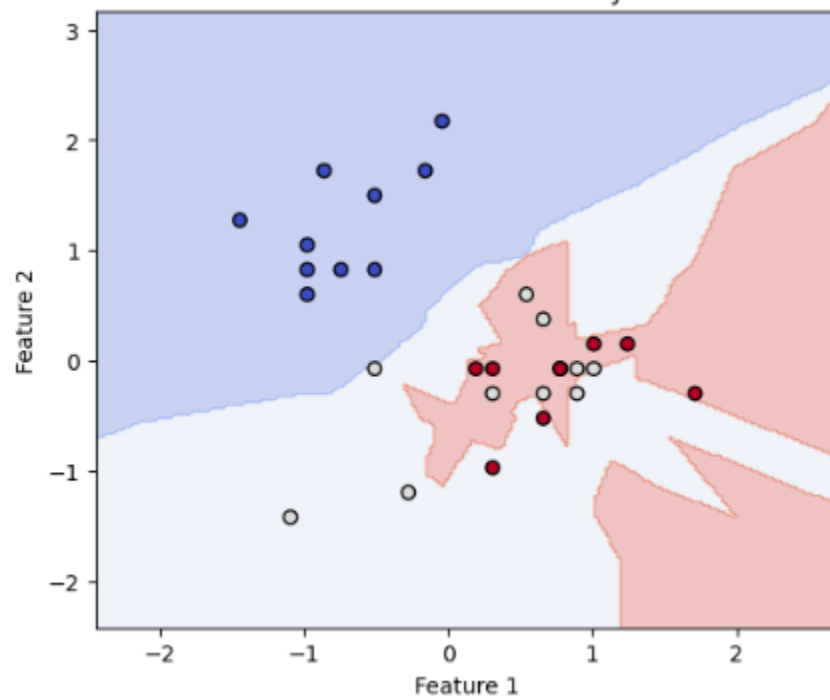


Figure 11: KNN Decision Boundary Graph

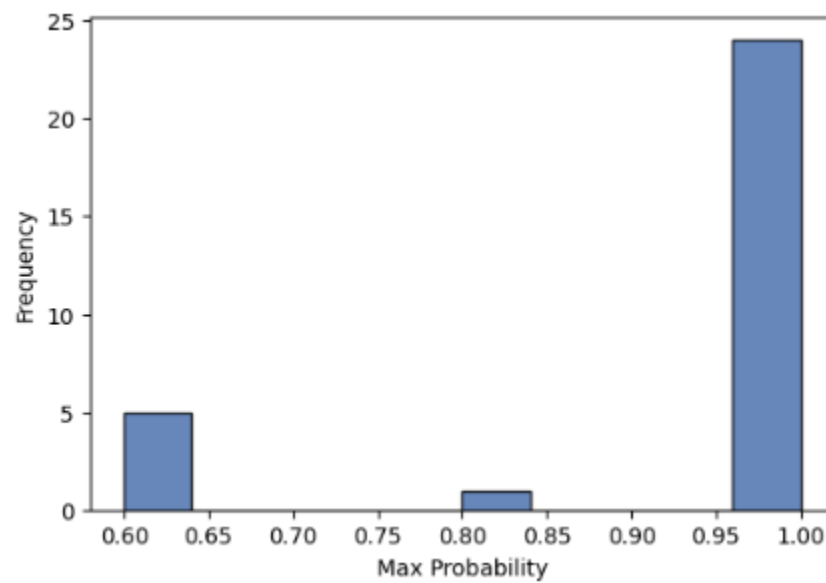


Figure 12: KNN Prediction Confidence Graph

IRIS FLOWER CLASSIFICATION

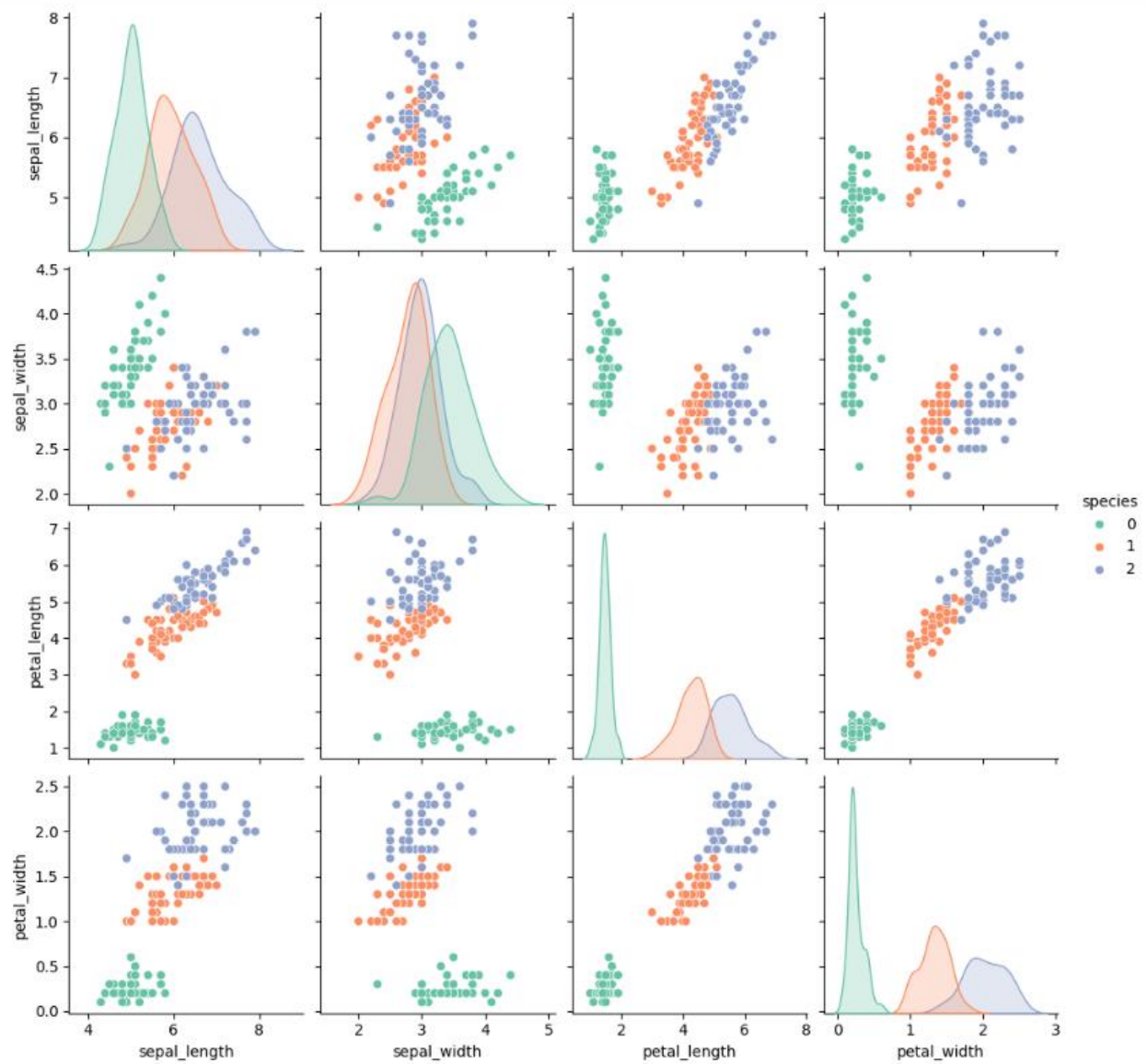


Figure 13: Pair Plot Visualization using KNN

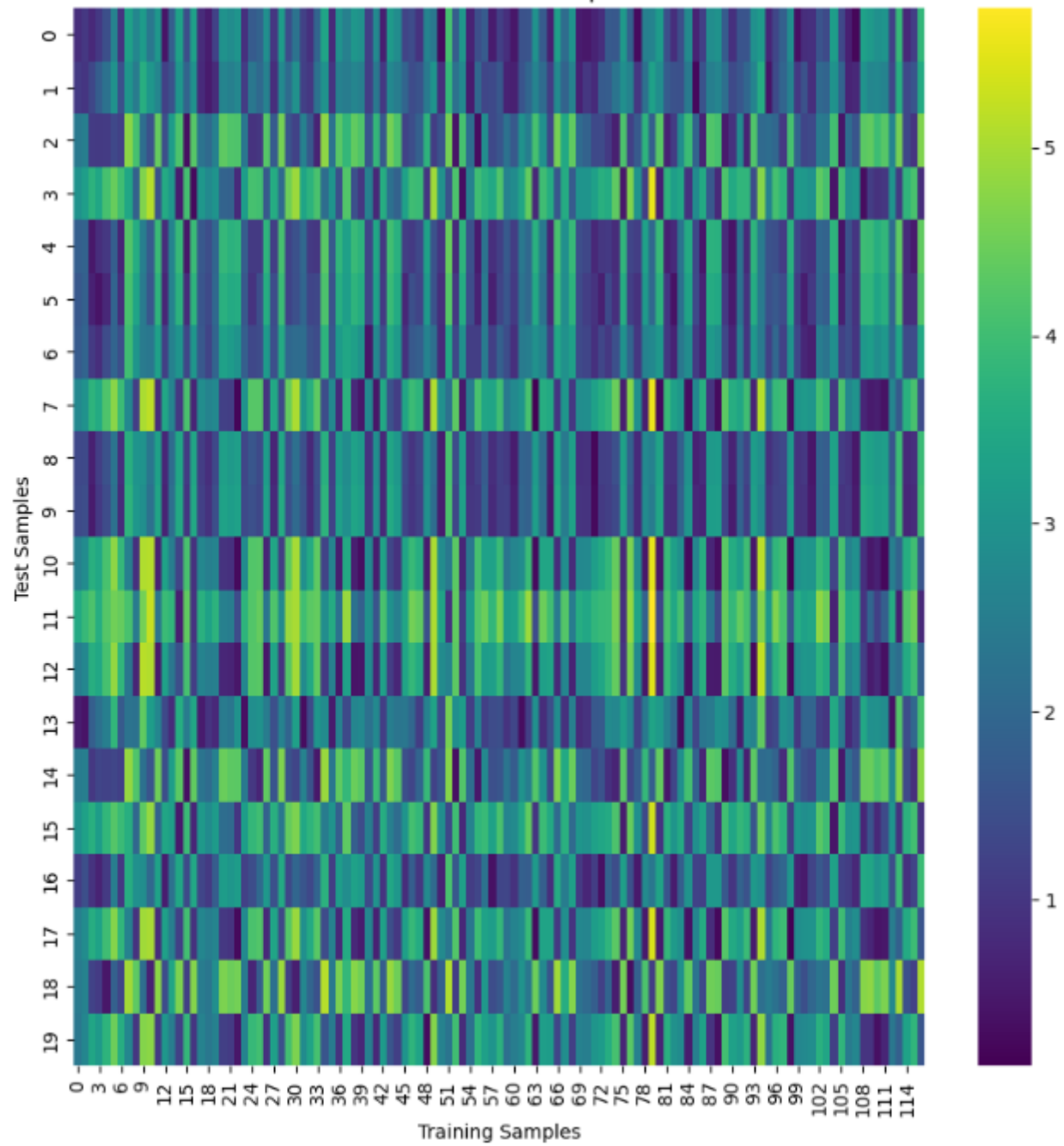


Figure 14: KNN, Distance Matrix (Sample Vs Train Values)

IRIS FLOWER CLASSIFICATION

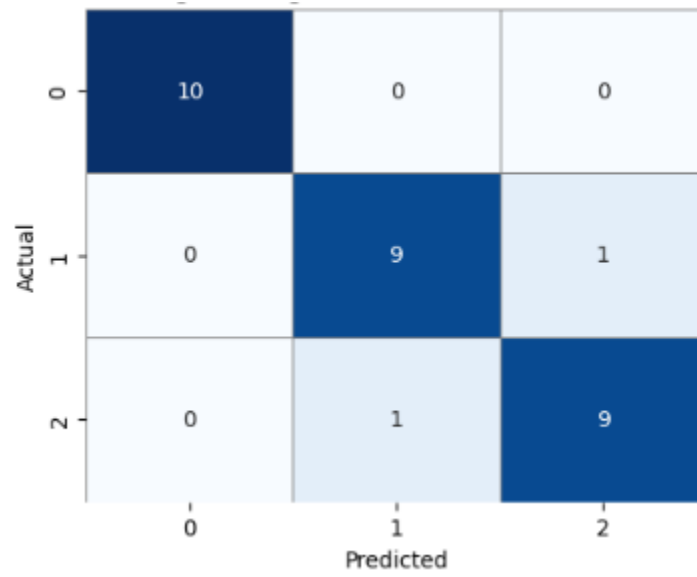


Figure 15: Confusion Matrix, Logistic Regression

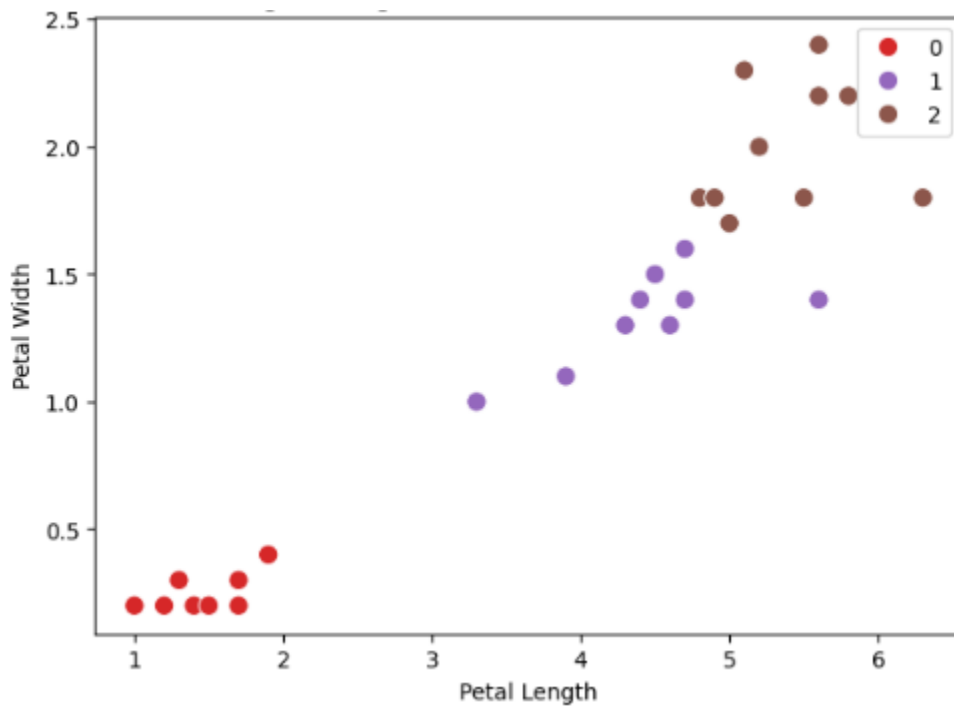


Figure 16: Prediction Scatter Plot, Logistic Regression

IRIS FLOWER CLASSIFICATION

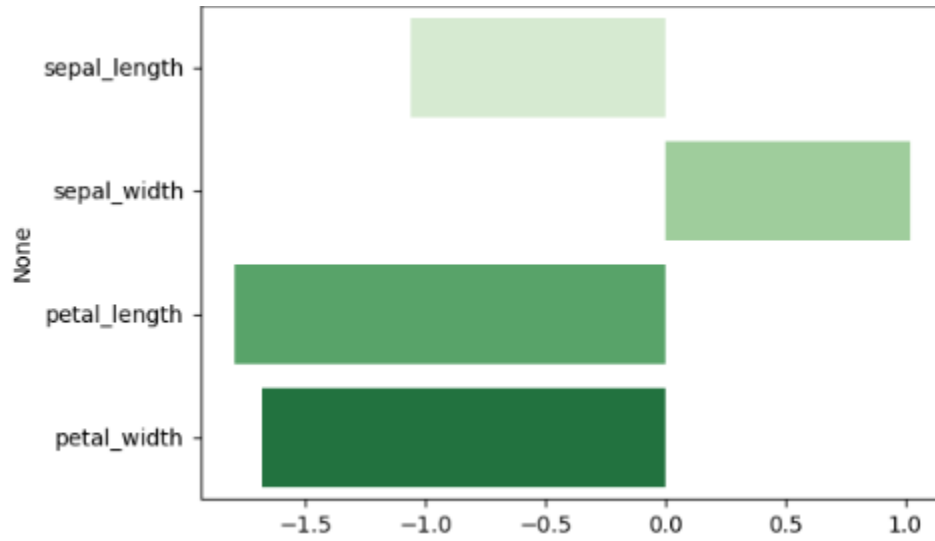


Figure 17: Feature Importance Graph, Logistic Regression

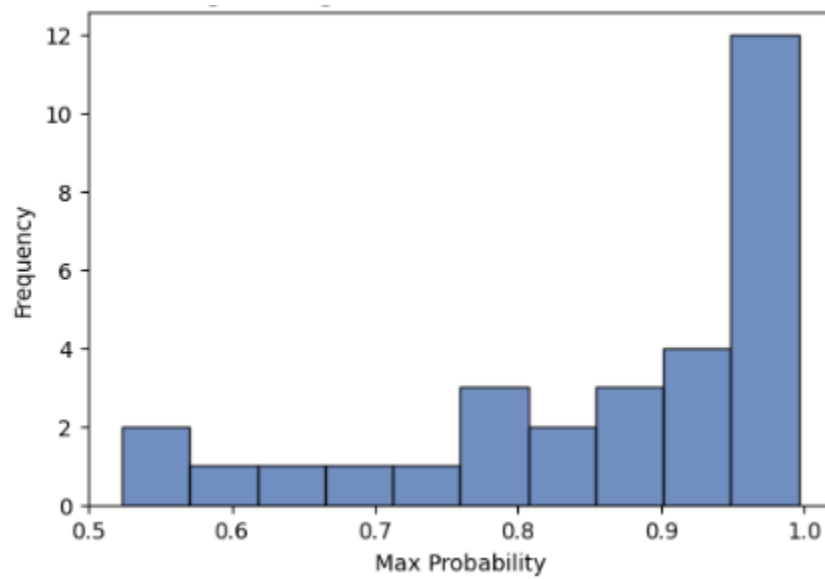


Figure 18: Logistic Regression, Prediction Confidence Graph

IRIS FLOWER CLASSIFICATION

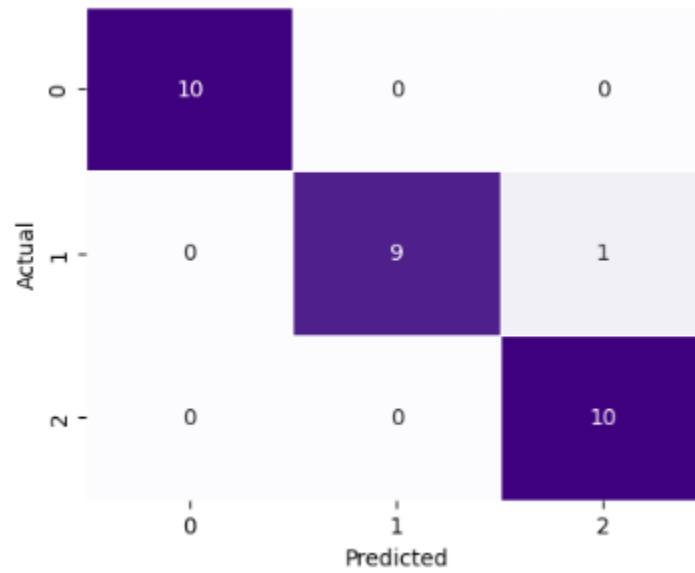


Figure 19: Confusion Matrix, Random Forest

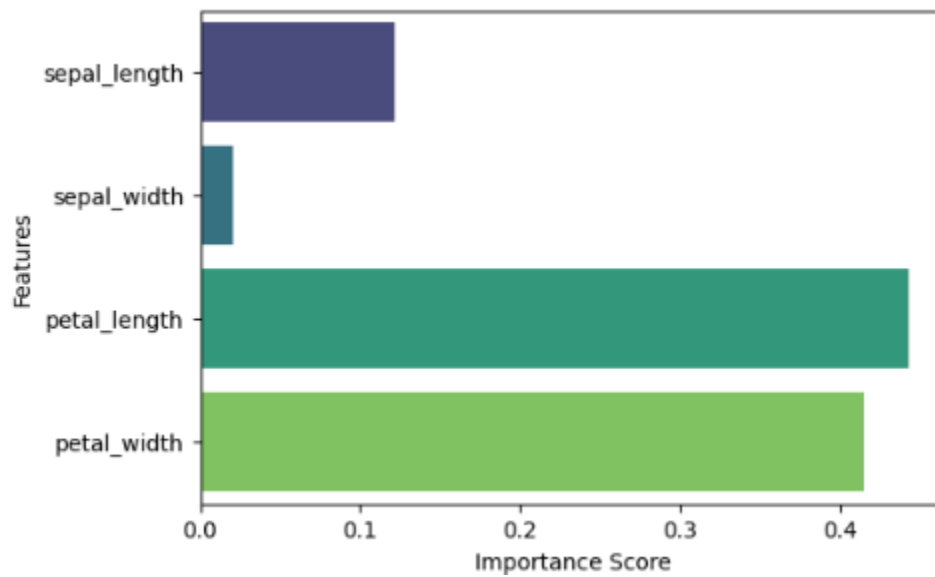


Figure 20: Feature Importance Graph, Random Forest

IRIS FLOWER CLASSIFICATION

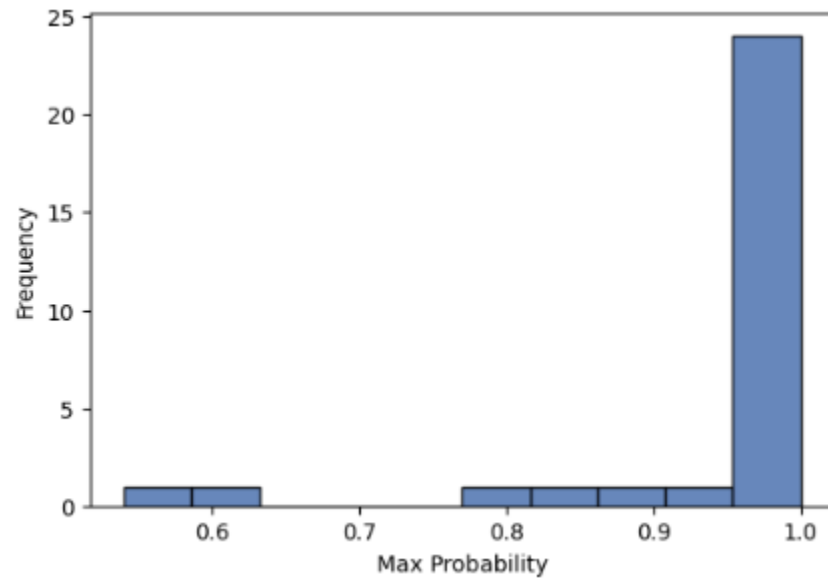


Figure 21: Random Forest, Prediction Confidence Graph

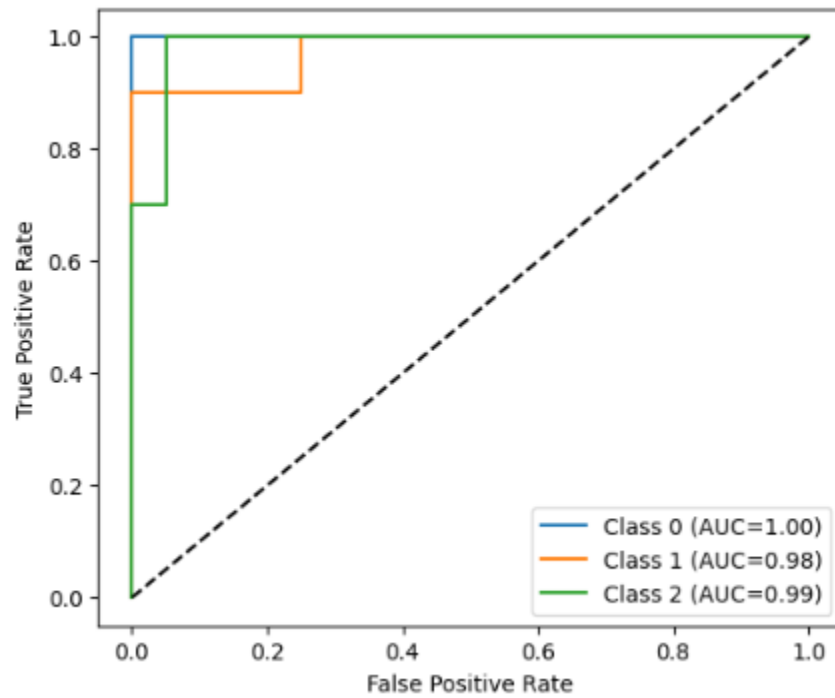


Figure 22: Random Forest, ROC Curve

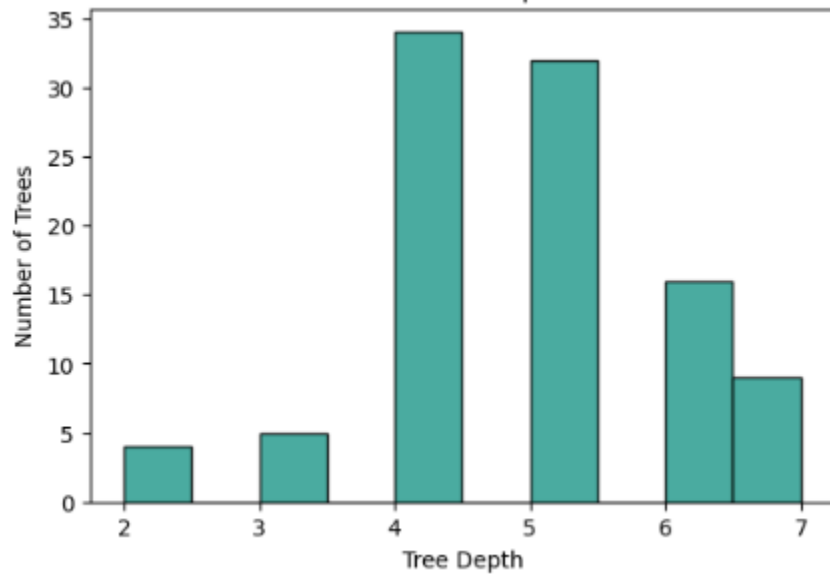


Figure 23: Random Forest, Tree Depth Distribution Graph

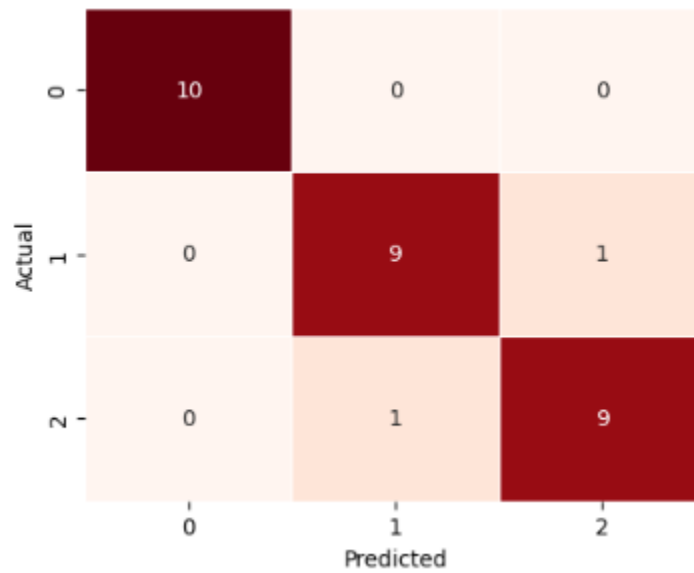


Figure 24: Confusion Matrix, Decision Tree

IRIS FLOWER CLASSIFICATION

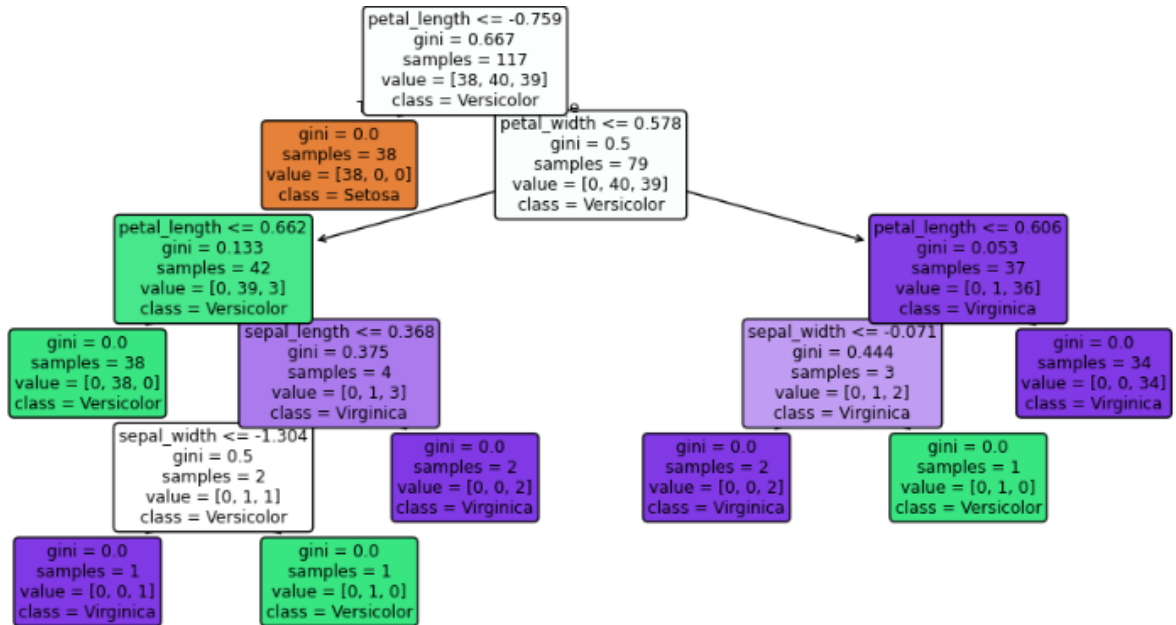


Figure 25: Sample Tree Structure, Decision Tree

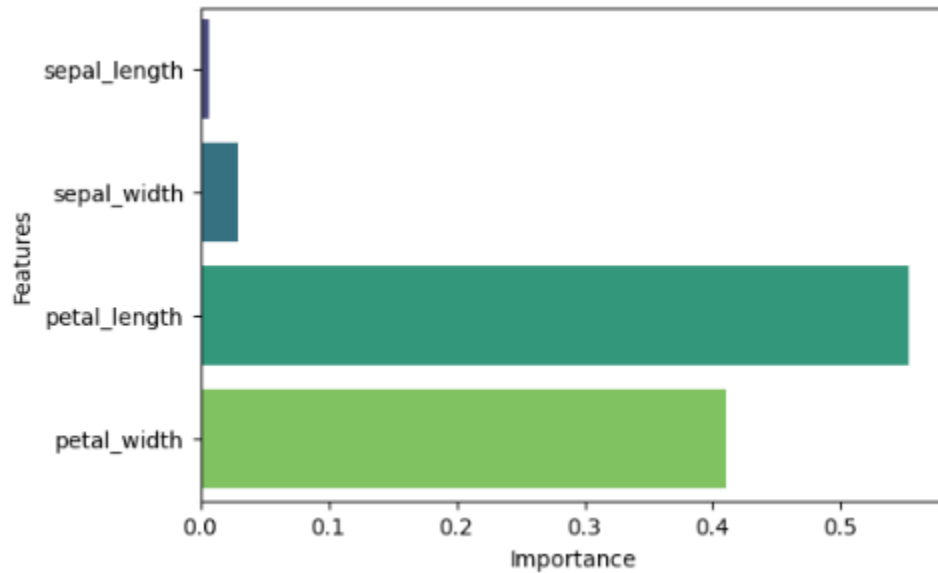


Figure 26: Feature Importance Graph, Decision Tree

IRIS FLOWER CLASSIFICATION

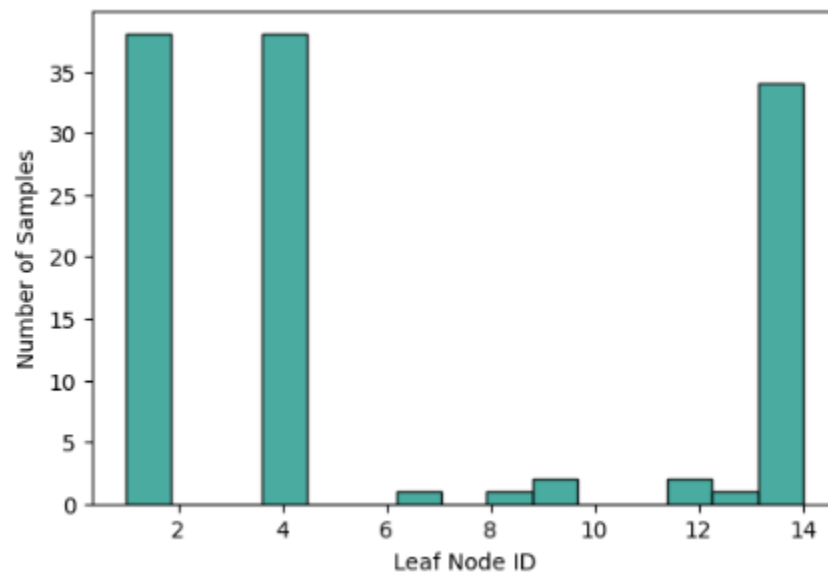


Figure 27: Leaf Node Distribution Graph, Decision Tree

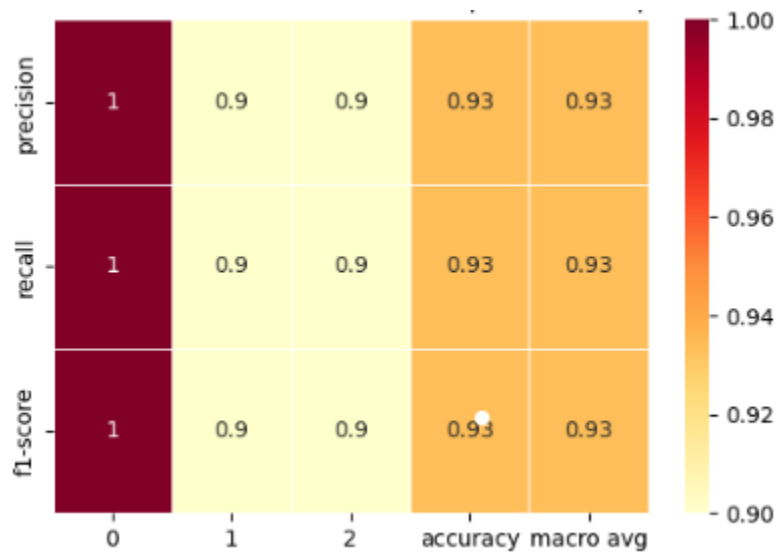


Figure 28: Classification Report Heatmap, Decision Tree

IRIS FLOWER CLASSIFICATION

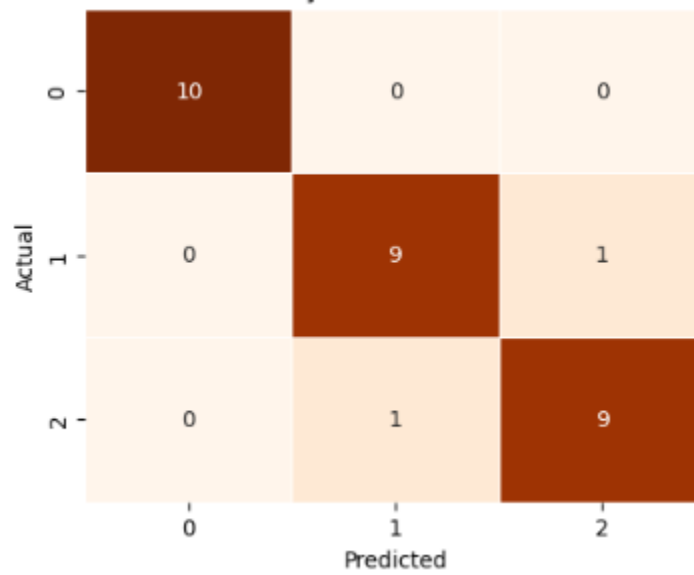


Figure 29: Naïve Bayes, Confusion Matrix

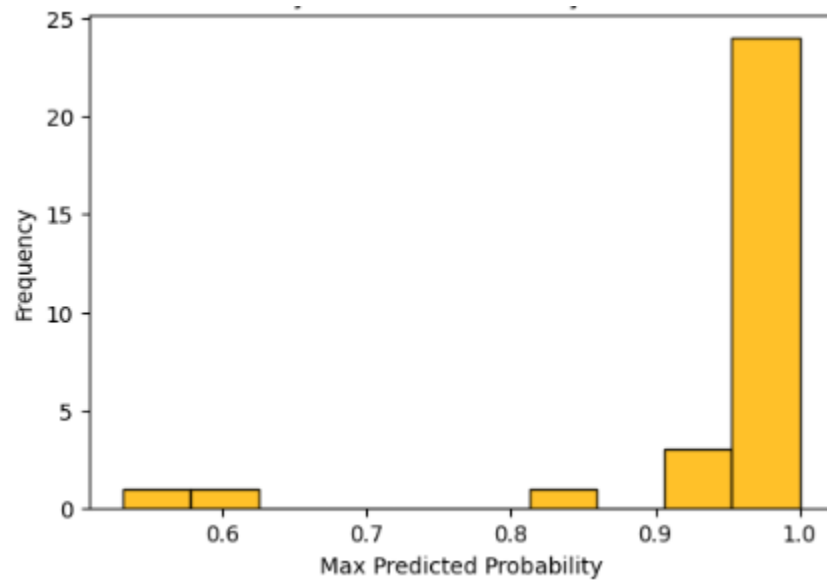


Figure 30: Naïve Bayes, Class Probability Distribution Graph

IRIS FLOWER CLASSIFICATION

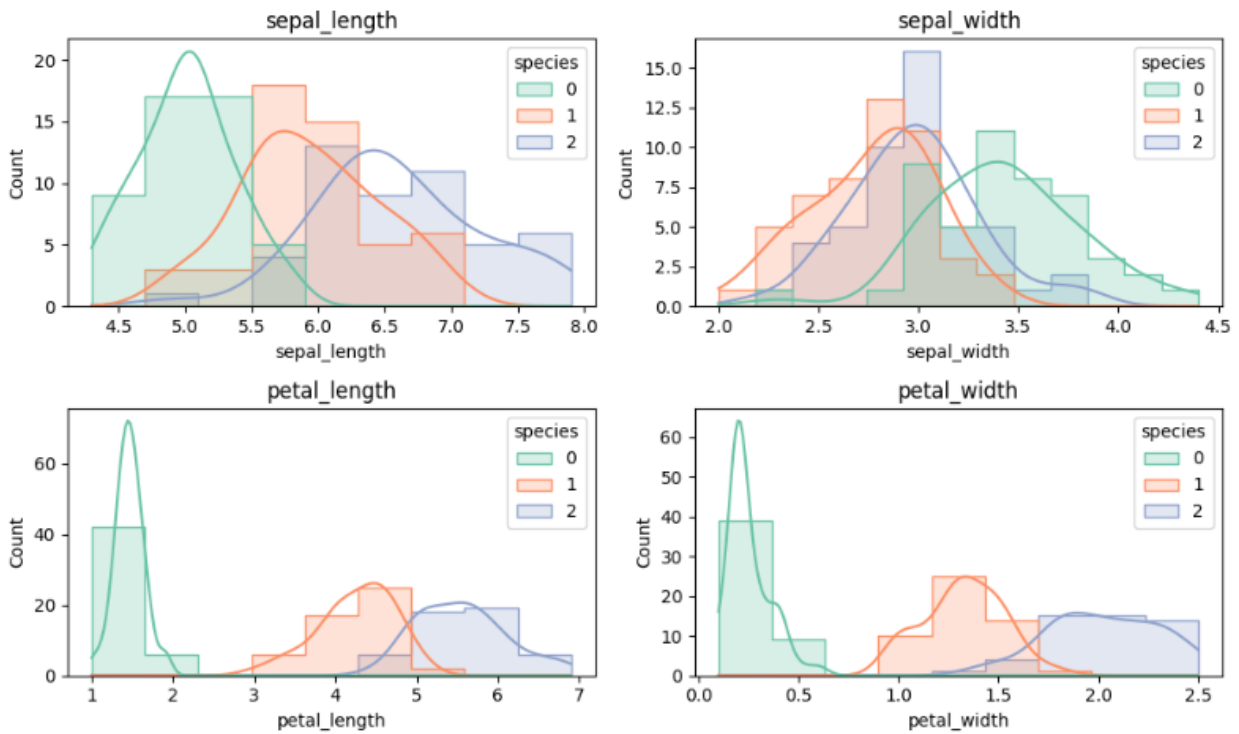


Figure 30: Naïve Bayes, Kernel Density Estimation Graph

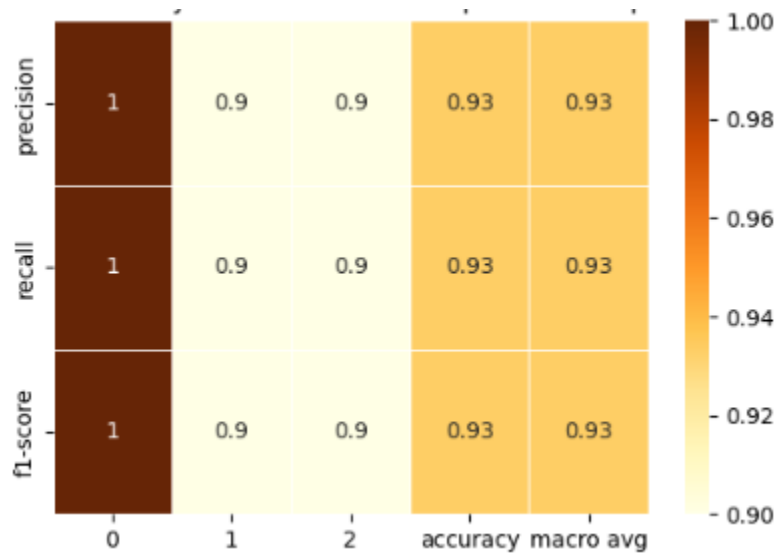


Figure 31: Naïve Bayes, Classification Report Heatmap

IRIS FLOWER CLASSIFICATION

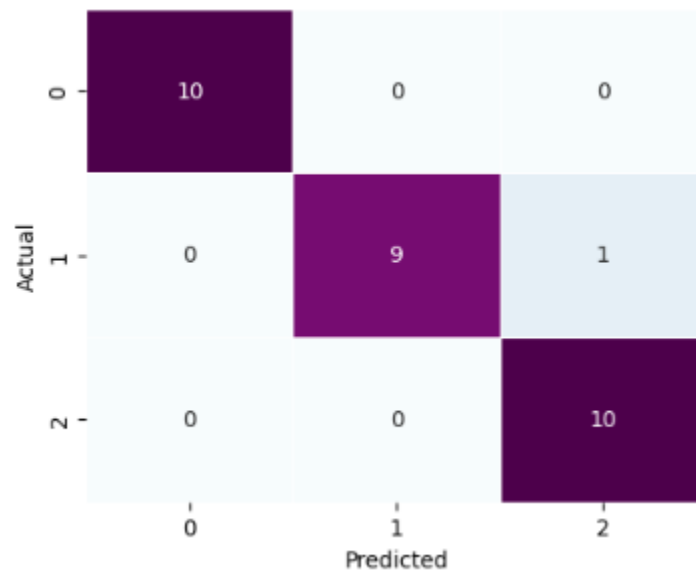


Figure 32: SVM, Confusion Matrix

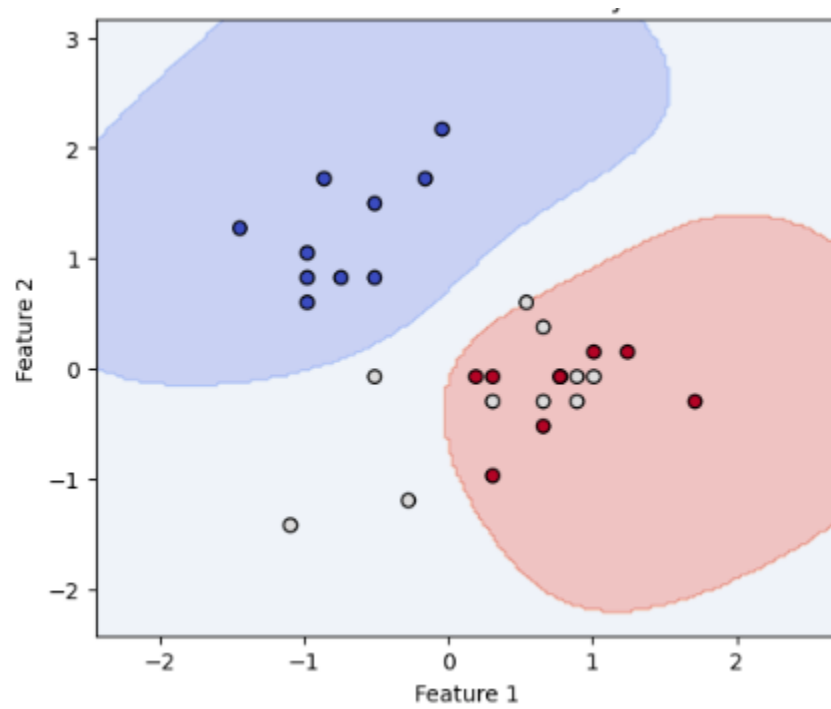


Figure 33: SVM, Decision Boundary

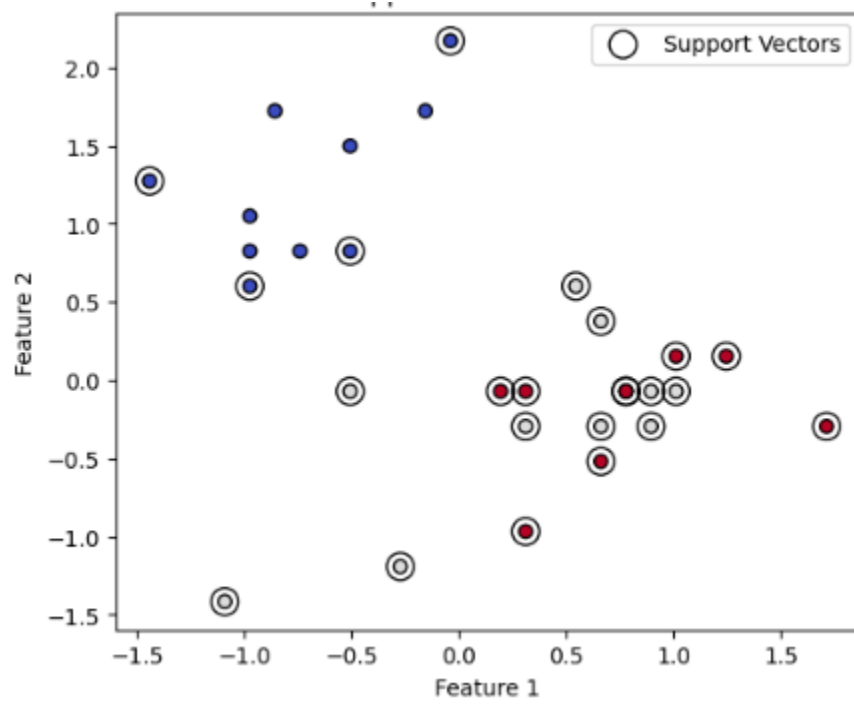


Figure 34: Support Vectors Visualization Graph

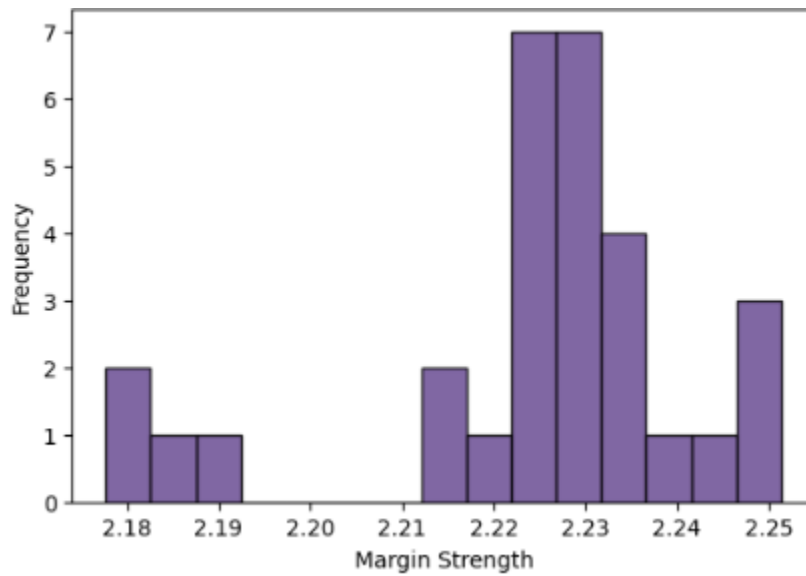


Figure 35: Decision Margin Distribution Graph, SVM

IRIS FLOWER CLASSIFICATION

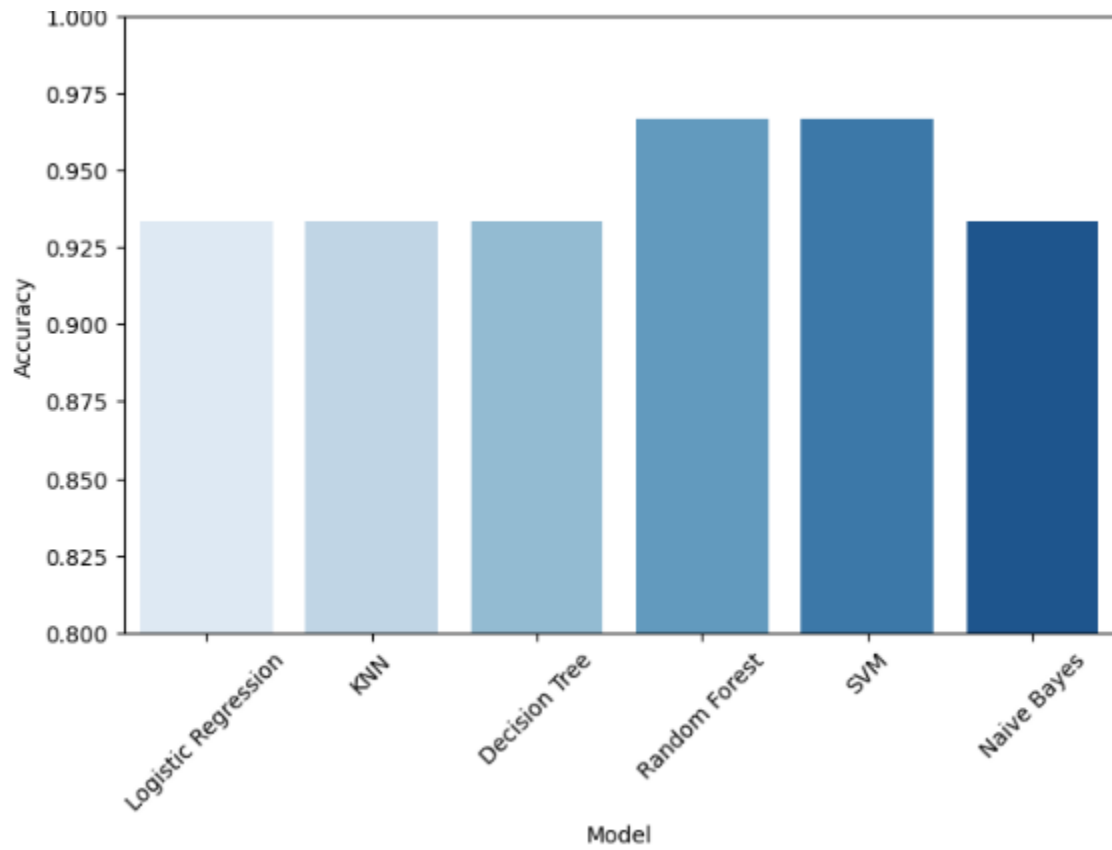


Figure 36: Accuracy Comparison Graph for All Models

IRIS FLOWER CLASSIFICATION

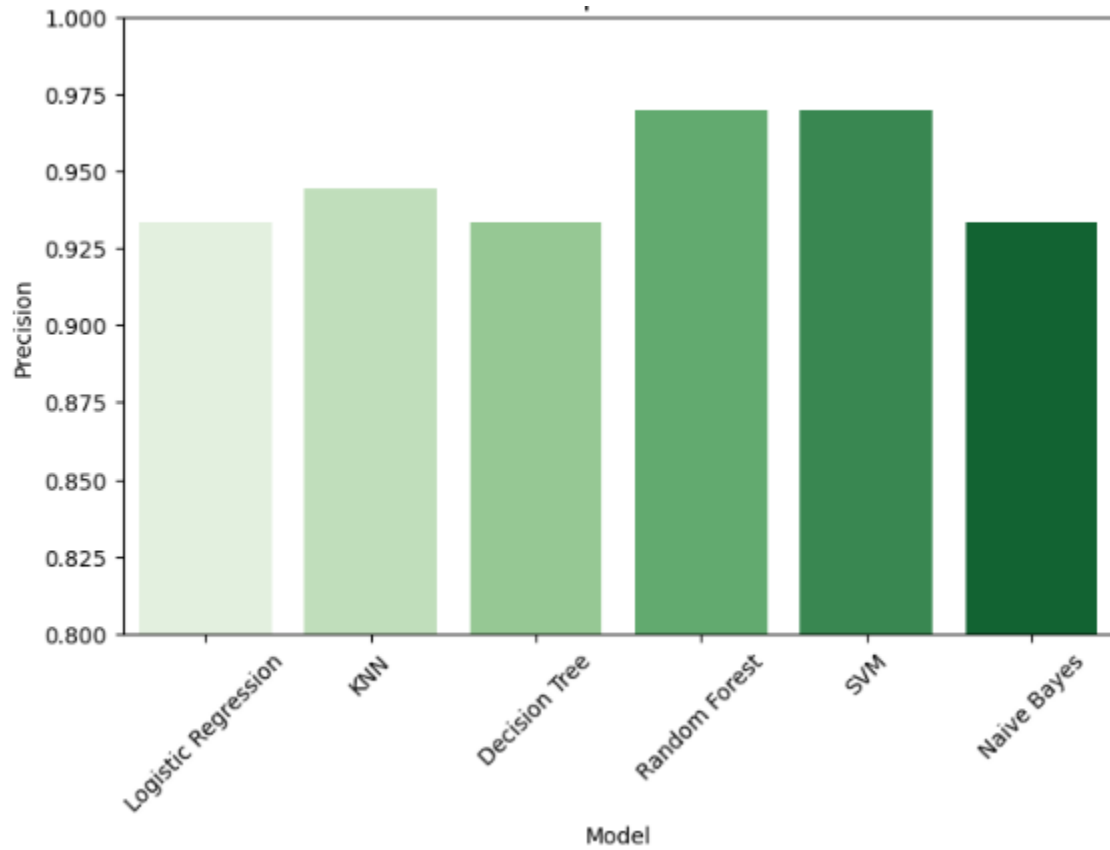


Figure 37: Precision Comparison Graph for All Models

IRIS FLOWER CLASSIFICATION

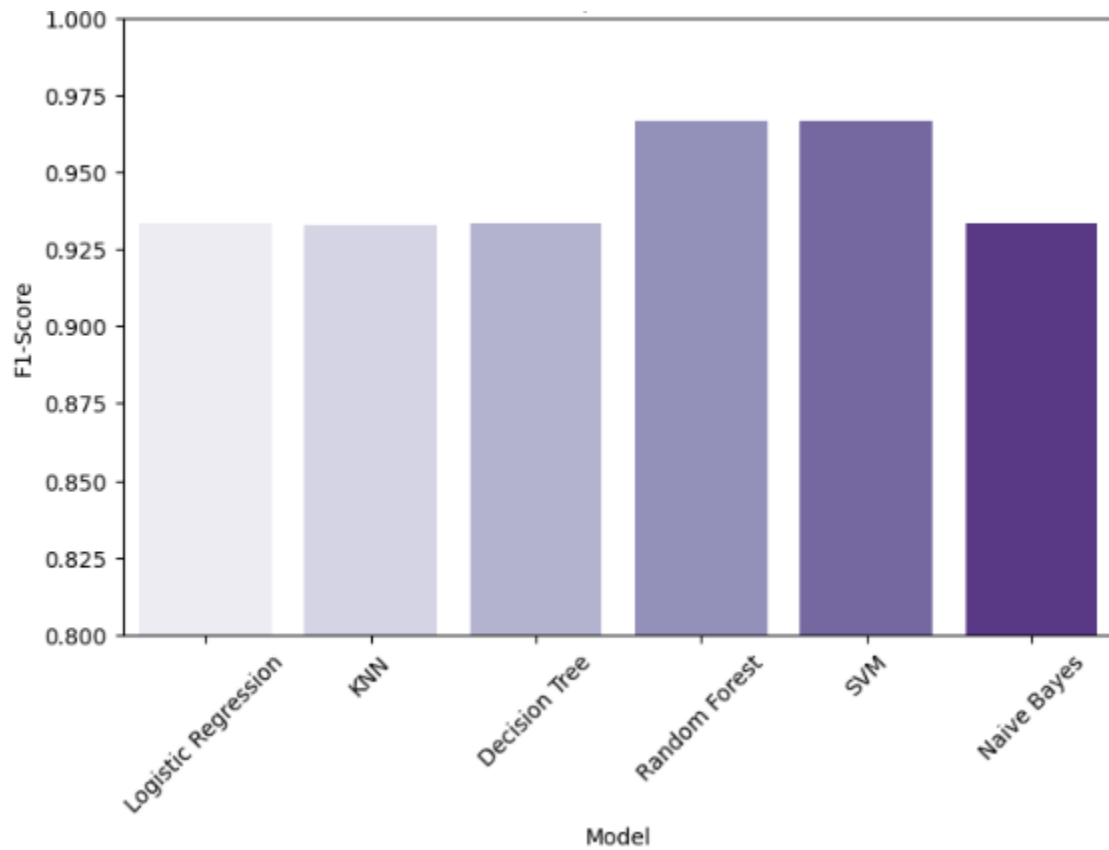


Figure 38: F1- Score Comparison Graph for All Models

IRIS FLOWER CLASSIFICATION

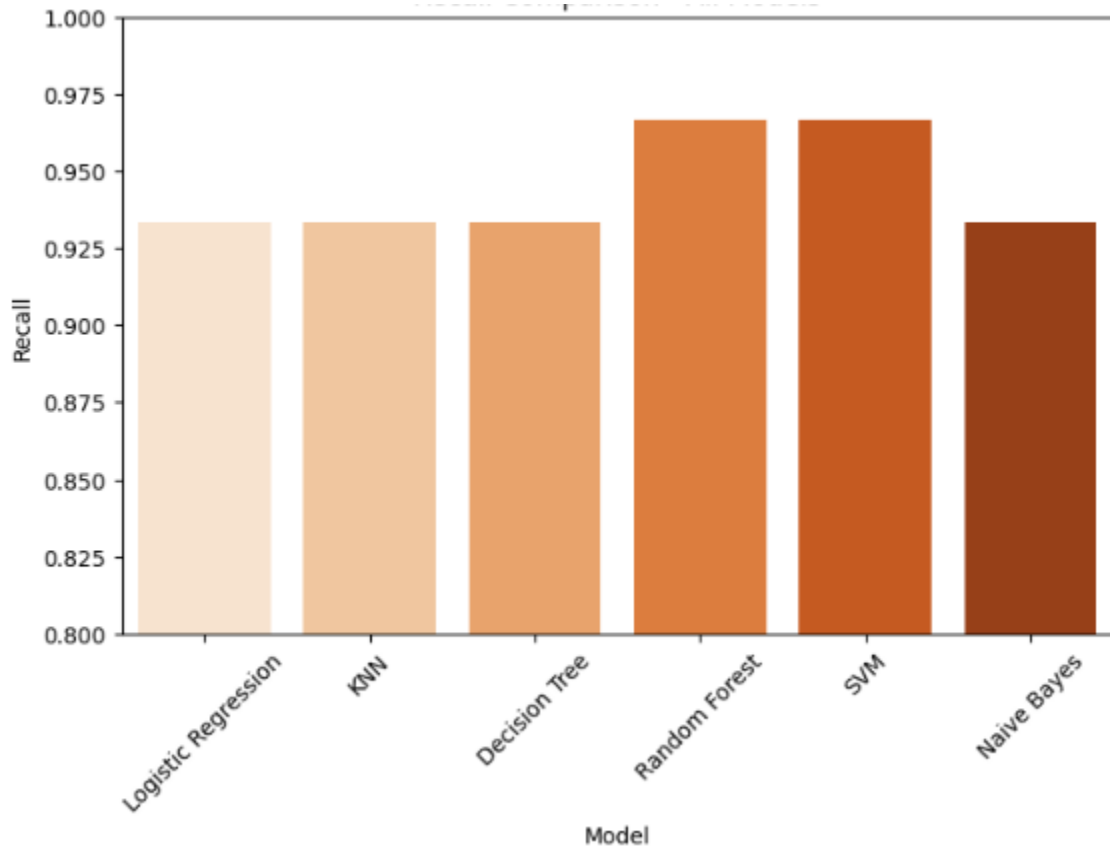


Figure 39: Recall Comparison Graph for All Models

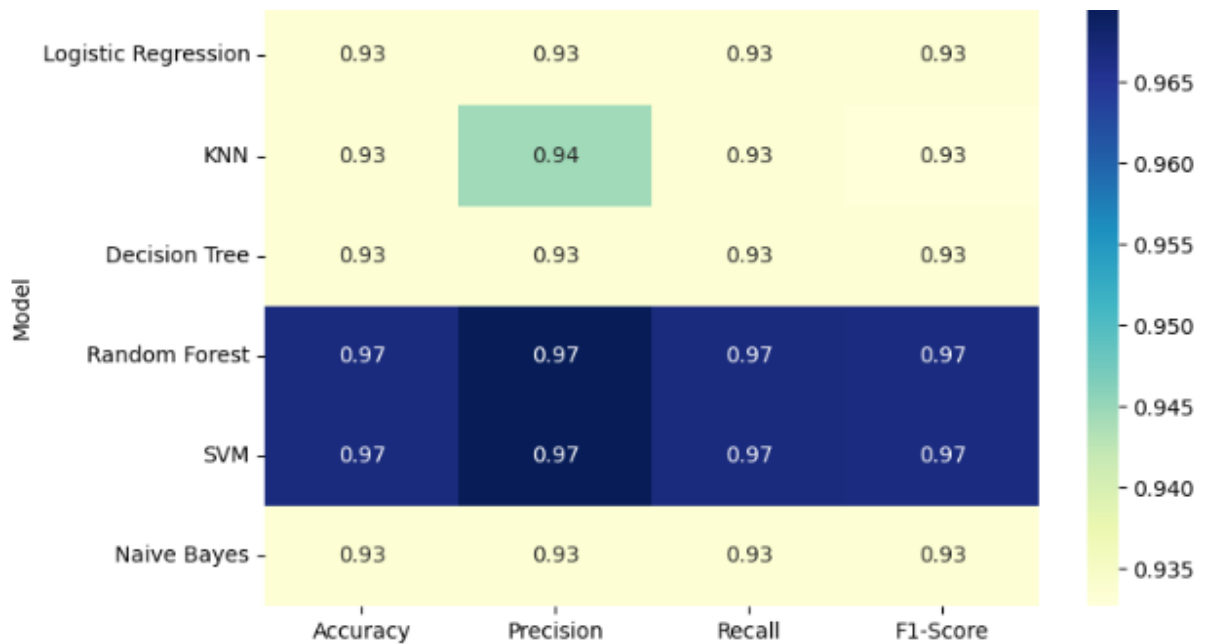


Figure 40: Models Metrics Heatmap

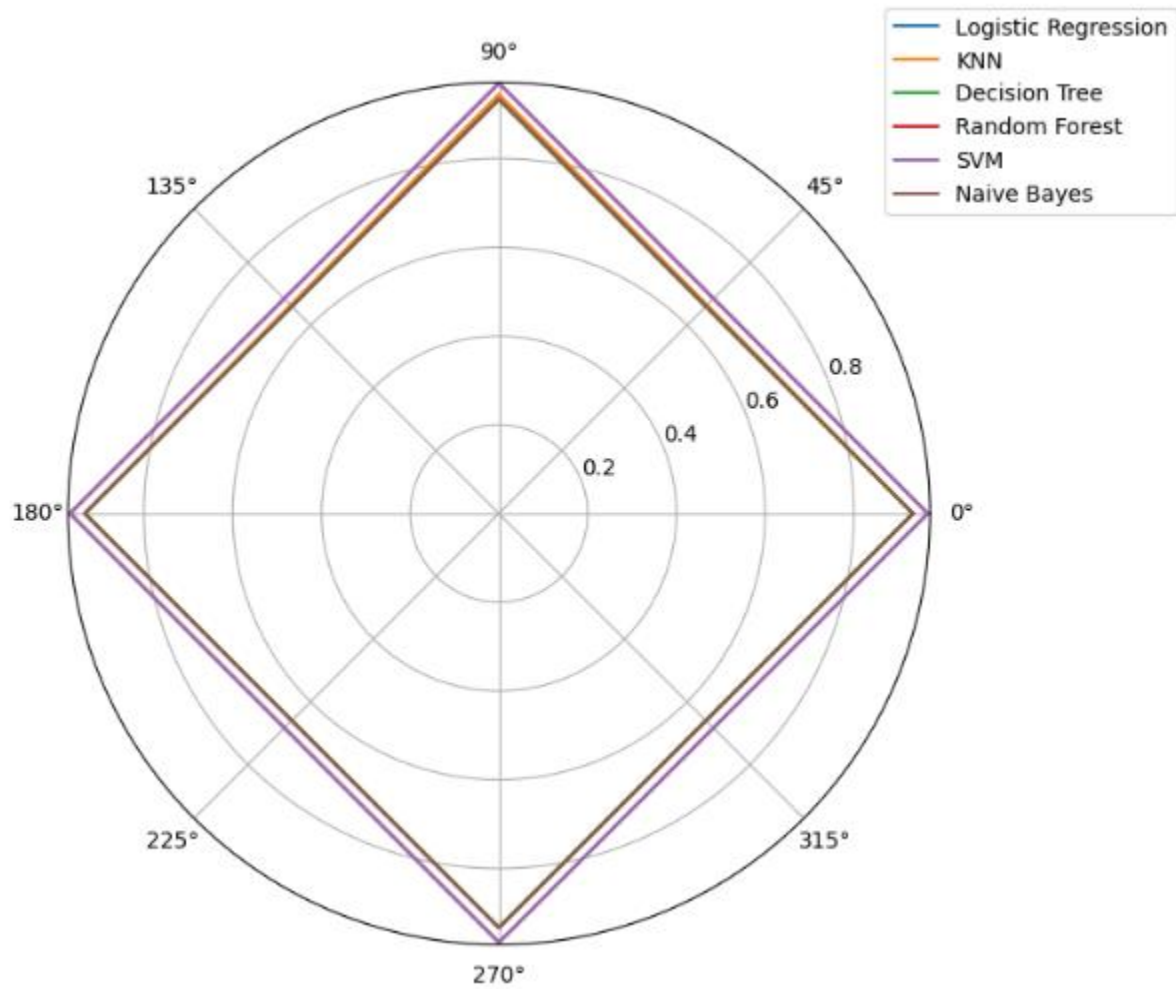


Figure 41: Models Comparison Radar Chart

IRIS FLOWER CLASSIFICATION

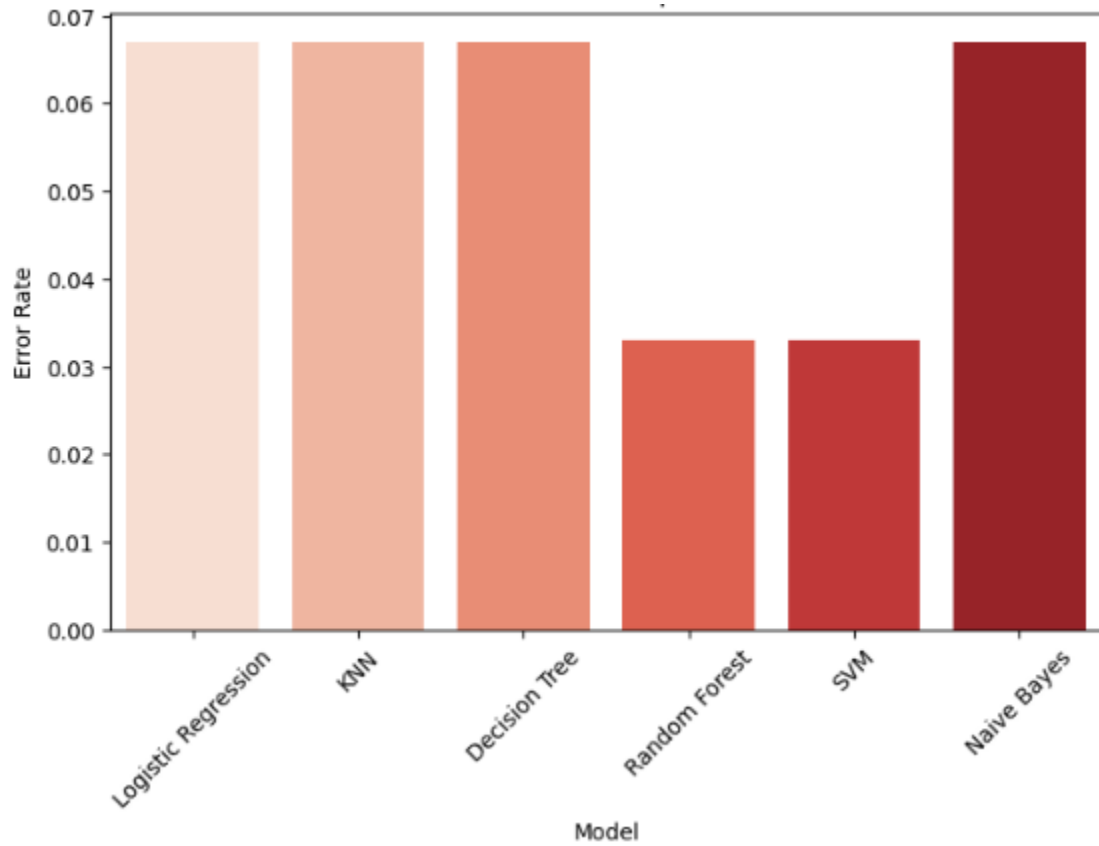


Figure 42: Error Rate Comparison Graph for All Models

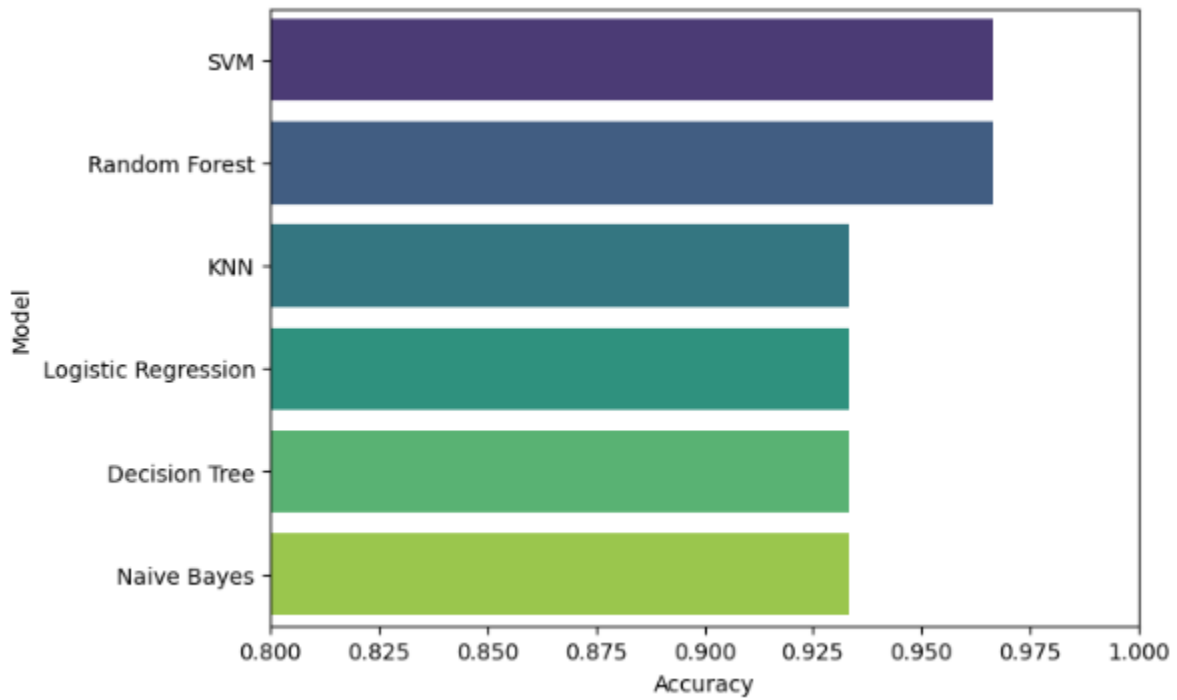


Figure 43: Models Ranking According to their Accuracy

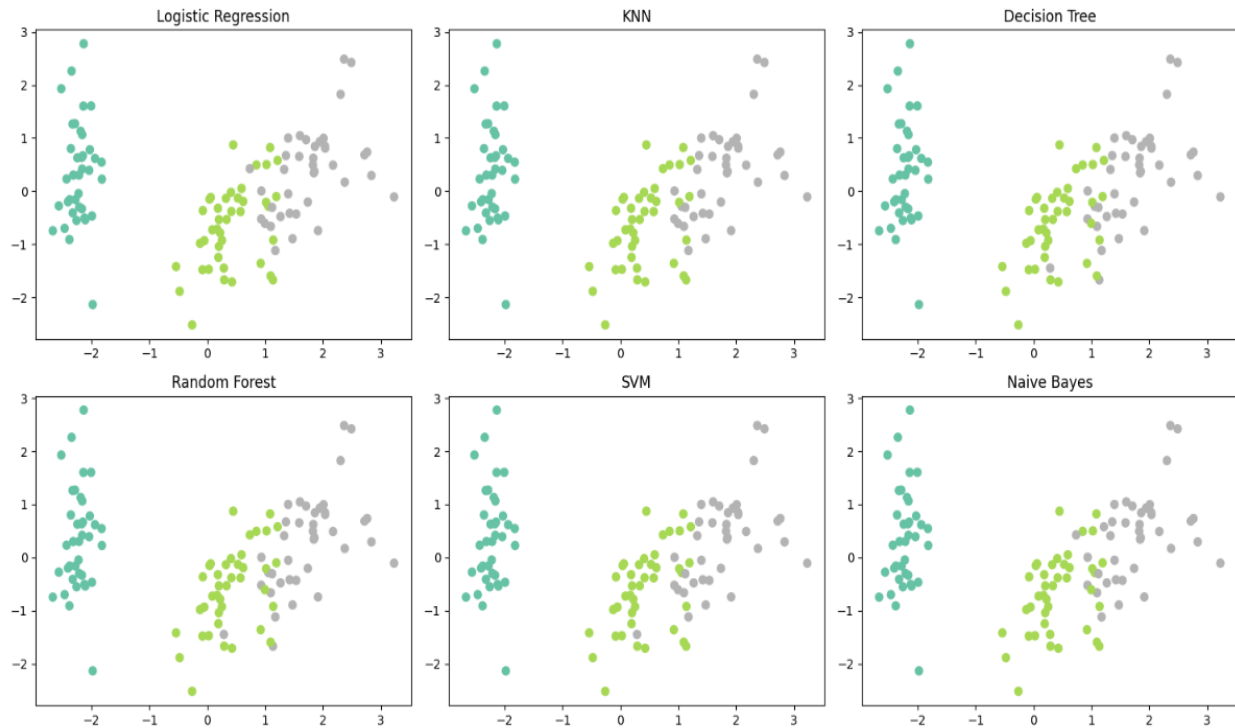


Figure 44: PCA Based Classification Visualization for All Models

```
Best Model Details:
Model          SVM
Accuracy       0.966667
Precision      0.969697
Recall         0.966667
F1-Score       0.966583
Error Rate     0.033333
Training Time  0.005579
Prediction Time 0.001004
Name: 4, dtype: object
```

Chart 6: Best Model Selection

```
Best Model: SVM
```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	1.00	0.90	0.95	10
Iris-virginica	0.91	1.00	0.95	10
accuracy			0.97	30
macro avg	0.97	0.97	0.97	30
weighted avg	0.97	0.97	0.97	30

Chart 7: Classification Report

	Actual Species	Predicted Species
0	Iris-versicolor	Iris-virginica
1	Iris-virginica	Iris-virginica
2	Iris-setosa	Iris-setosa
3	Iris-setosa	Iris-setosa
4	Iris-virginica	Iris-virginica
5	Iris-setosa	Iris-setosa
6	Iris-virginica	Iris-virginica
7	Iris-versicolor	Iris-versicolor
8	Iris-virginica	Iris-virginica
9	Iris-virginica	Iris-virginica
10	Iris-versicolor	Iris-versicolor
11	Iris-versicolor	Iris-versicolor
12	Iris-versicolor	Iris-versicolor
13	Iris-versicolor	Iris-versicolor
14	Iris-setosa	Iris-setosa

Chart 8: 15 Random samples, Actual Vs Predicted Results

```

Iris Species Predictions
Enter Sepal Length: 5.1
Enter Sepal Width: 3.5
Enter Petal Length: 1.4
Enter Petal Width: 0.2
Predicted Iris Species: Iris-setosa
    
```

Chart 9: User Input Test case for Iris- Setosa

```

Iris Species Predictions
Enter Sepal Length: 6.0
Enter Sepal Width: 2.9
Enter Petal Length: 4.5
Enter Petal Width: 1.5
Predicted Iris Species: Iris-versicolor
    
```

Chart 9: User Input Test case for Iris- Versicolor

IRIS FLOWER CLASSIFICATION

Iris Species Predictions
Enter Sepal Length: 6.8
Enter Sepal Width: 3.0
Enter Petal Length: 5.5
Enter Petal Width: 2.1
Predicted Iris Species: Iris-virginica

Chart 9: User Input Test case for Iris- Virginica

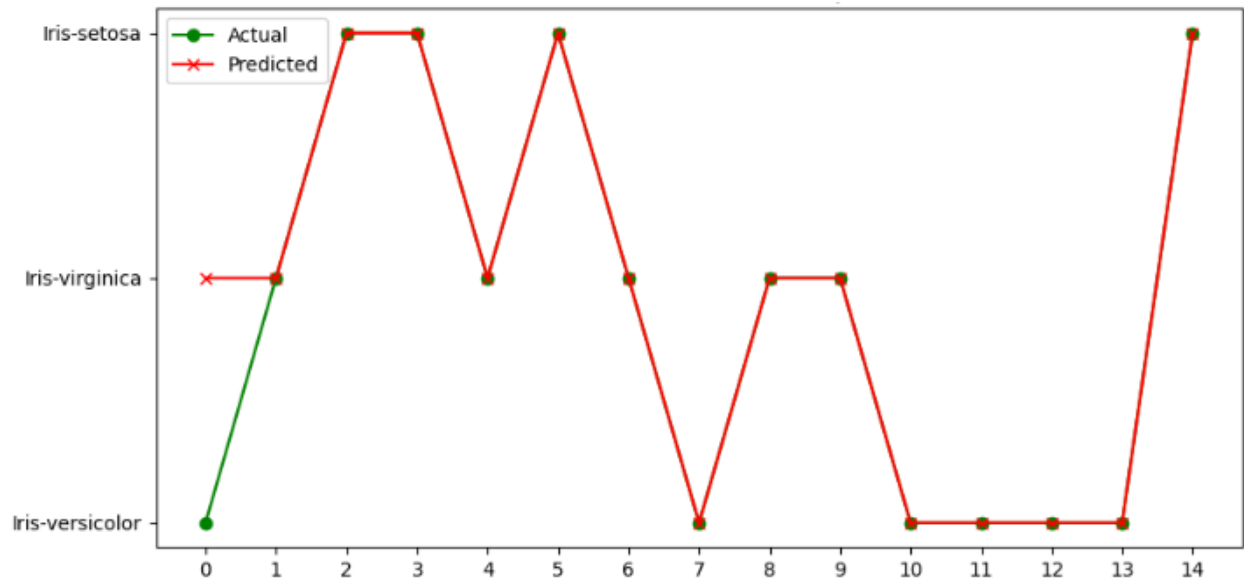


Figure 45: Actual Vs Predicted Samples Results

Conclusion

The Iris Species Classification System was successfully developed using multiple machine learning algorithms to accurately predict the species of Iris flowers based on their physical measurements. The project included all major steps of a complete machine learning pipeline, from data preprocessing and exploratory data analysis to model training, evaluation, and final selection.

1. The dataset was clean, balanced, and well-structured, which made it suitable for classification tasks without requiring heavy preprocessing.
2. Exploratory Data Analysis showed that petal features, especially petal length and petal width, provide strong separation between species and are the most important predictors.
3. Feature engineering techniques such as Label Encoding and StandardScaler improved data compatibility and ensured better model performance.
4. Multiple machine learning models were trained, including Logistic Regression, KNN, Decision Tree, Random Forest, SVM, and Naive Bayes.
5. Among all models, **Random Forest and SVM performed the best**, achieving the highest accuracy of 0.9667 with strong and balanced evaluation metrics.
6. **SVM** was selected as the final model due to its strong generalization ability and consistent performance across all evaluation measures.
7. The final system is capable of accurately classifying Iris species and demonstrates the effectiveness of machine learning in solving real-world classification problems.

Overall, this project highlights how proper data analysis, feature selection, and model evaluation can lead to highly accurate predictive systems.

Writer's Introduction



Assalam-u-Alaikum,

I am **Muhammad Anas Rathore**, a dedicated student currently pursuing my undergraduate studies in Computer Science. I maintain a strong academic record with a CGPA of **3.9** and continuously strive to expand my technical knowledge while applying innovative technologies to solve real-world challenges.

My primary areas of interest include **Artificial Intelligence, Machine Learning, Deep Learning, Data Science, and Software Development**. Throughout my academic journey, I have worked on multiple practical projects that have strengthened my technical expertise and enhanced my problem-solving abilities. These experiences have enabled me to bridge the gap between theoretical concepts and real-world applications.

In addition to my academic projects, I have gained valuable hands-on experience through internships and collaborative development work, where I contributed to software and AI-based solutions. These experiences have helped me strengthen my technical, analytical, and communication skills while working in professional environments.

As a lifelong learner, I remain committed to exploring emerging technologies, enhancing my skill set, and contributing to impactful innovations in the fields of Artificial Intelligence and Software Engineering.

A handwritten signature in black ink, appearing to read 'Muhammad Anas Rathore', with a stylized flourish at the end.

Muhammad Anas Rathore